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Author(s): Frederico Belo, Xiaoji Lin and Fan Yang

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External Equity Financing Shocks, Financial Flows, and Asset Prices

Frederico Belo

INSEAD, University of Minnesota, and National Bureau of Economic Research

Xiaoji Lin

Carlson School of Management, University of Minnesota

Fan Yang

School of Business, University of Connecticut

We develop a dynamic model with time variation in external equity financing costs and show that variation in these costs is important for the model to quantitatively capture the joint dynamics of firms' asset prices, real quantities, and financial flows in the U.S. economy. Growth firms and high investment firms are less risky in equilibrium, because they can substitute more easily debt financing for equity financing when it becomes more costly to raise external equity, which are high marginal utility states. Using a model-implied proxy of aggregate equity issuance cost shocks, we provide empirical support for the model's economic mechanism. (*JEL* D53, E22, G12, G32)

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Corporations face substantial uncertainties about financing conditions. In particular, firms' ability to raise external funds varies over time with macroeconomic conditions. In this paper, we study the impact of time variation in the cost of issuing equity—a particular form of financial shocks—on asset prices, real quantities, and financial flows in the cross-section of U.S. publicly traded firms. We show both theoretically and empirically that variation in the firm's ability to raise external equity financing has a significant impact on equilibrium risk premiums in the cross-section. The economic mechanism emphasizes that the firm's ability to substitute between different marginal sources of external financing (equity and debt) during bad economic times is an important determinant of equilibrium risk premiums.

We develop a dynamic model economy with a large cross-section of firms that invest in physical capital, issue equity and debt or save using cash, to maximize the value of the firm for existing shareholders. Cross-sectional heterogeneity is driven by idiosyncratic productivity shocks. The key features of the model are (1) equity issuance is costly and this cost is stochastic, varying over time due to an aggregate equity issuance cost shock (henceforth equity ICS), and (2) debt issuance is limited by a standard collateral constraint. Consistent with Bolton, Chen, and Wang (2013) and Eisfeldt and Muir (2016) and additional empirical evidence reported here, the equity ICS acts as an additional source of aggregate economic fluctuations that is independent of aggregate productivity shocks, and is correlated with investor's marginal utility (stochastic discount factor). In particular, increases in the aggregate cost of issuing equity are associated with high marginal utility periods, that is, bad economic times.

We interpret the equity ICS as a financial shock because of its direct impact on firms' ability to raise external equity financing, but this shock has a broad interpretation. In essence, the equity ICS captures a time-varying wedge between the fundamental value of the equity for managers and the value that outside investors are willing to pay for new equity. This wedge can vary due to changes in information asymmetries or agency frictions that have a systematic aggregate impact on investors' valuation of new equity. Similarly, the wedge may vary because of an increase (decrease) in investors' willingness to supply equity capital to firms (e.g., because of changes in investor sentiment, or changes in aggregate liquidity), thus making it effectively less (more) costly for firms to raise new equity capital. We discuss in more detail the micro-foundations of time-varying equity issuance costs in Section 1.

The model generates cross-sectional variation in expected stock returns across book-to-market (value premium) and investment rate (investment return spread) portfolios that is similar to that observed in the data. This variation arises endogenously in the model due to the interaction between firms' productivity, investment, equity ICS, and the collateral constraint. The economic mechanism driving the results relies on the time variation in firms' ability to substitute between different marginal sources of external financing (equity and debt) during bad economic times (high marginal utility states). Because growth firms

and high investment firms can substitute more easily debt financing for equity financing when it becomes more costly to raise external equity, which are high marginal utility states, these firms are less risky in equilibrium.

The exact economic mechanism in the model operates as follows. Firms with high idiosyncratic productivity are expanding firms with high investment demand. When a negative equity ICS hits the economy, it becomes more difficult for all firms to raise external equity. But the high productivity firms can still finance investment through debt because their collateral constraint is less binding, and hence are still able to increase their future dividend payout allowing its continuation value to increase. As a result, these firms are relatively less affected by the equity ICS, and hence their returns covary less with the equity ICS. Therefore, these firms have relatively lower risk and hence lower expected stock returns in equilibrium.

Compared with the firms with high idiosyncratic productivity, the firms with low idiosyncratic productivity are relatively more affected by the negative equity ICS. These firms are experiencing a decrease in their productivity causing sales and profits to fall. These firms want to downsize, and hence the capital stock of these firms is shrinking. Because of the decrease in internal funds (sales), and more important, of the increase in the cost of external equity financing (they would otherwise be able to raise external equity to pay off debt if it was not costly to access the equity market), the low productivity firms de-leverage to avoid hitting the binding collateral constraint as their collateral value falls. Their dividend payout falls below the steady-state level for a long time, and their continuation value falls. As a result, these firms are relatively more affected by the equity ICS and hence their returns covary more with this aggregate shock. Therefore, these firms have relatively higher risk and hence higher expected stock returns in equilibrium.

In the model, and consistent with the data (see, e.g., İmrohoroglu and Tüzel 2014), the high productivity firms tend to be growth firms and high investment firms. Thus, the model generates sizeable value and investment return spreads consistent with the data. In addition, because the average return spreads in the model are driven by the endogenous differential exposure of the returns of the portfolios to the aggregate equity ICS, and not by differential exposure to the market factor (which is mostly driven by the aggregate productivity shock), the model also replicates the failure of the standard capital asset pricing model (CAPM) in explaining the cross-sectional variation in the expected returns of the portfolios considered here.

Empirically, following Eisfeldt and Muir (2016), we use the structure of the model to estimate an equity ICS proxy in the U.S. economy from 1964 to 2013, and test the models' assumptions and economic mechanisms. This estimation is required because, as noted, a substantial part of the equity issuance costs are due to, for example, asymmetric information and agency frictions, which are not directly observed in the data. In the model, variation in two key moments, the fraction of firms' issuing equity in the cross-section and the

aggregate debt-to-physical capital ratio, when combined using a multivariate regression, capture almost all the variation in the true equity ICS in the model (the correlation between the fitted value of the regression and the true equity ICS is 93%). Intuitively, the first moment captures changes in firms' extensive margin of external equity issuance when the cost of issuing equity varies, while the second moment captures the substitution effect between debt and equity when the relative cost of these sources of external financing changes. We then apply the model-implied relationship between the true equity ICS and the analogous empirical moments in the data to construct a time series of the equity ICS in the real data.

We show that the equity ICS proxy captures systematic risk in the economy. Controlling for the aggregate market factor, firms' differential exposure to the equity ICS proxy helps explain cross-sectional variation in average stock returns of portfolios sorted by book-to-market, investment rate, and other portfolios. In addition, consistent with the model's assumption, we document that investors require a higher risk premium for holding assets that are more positively exposed to the equity ICS proxy, that is, assets that do poorly when it is more costly to issue equity (low equity ICS states). Because value firms and low investment firms have a relatively higher exposure (covariance) to the equity ICS proxy, these firms are riskier and thus, have higher returns in equilibrium. Augmenting the CAPM with equity ICS proxy as an additional factor, we show that the two-factor model significantly outperforms the CAPM in pricing the cross-section of stock returns of these portfolios.

Furthermore, we provide empirical support for the model's debt-equity substitution economic mechanism. In the data, more productive (growth, high investment) firms reduce the use of equity financing and increase the use of debt financing as the cost of issuing equity increases. That is, these firms substitute equity financing for debt financing when external equity financing is too costly. In contrast, the less productive (value, low investment) firms reduce the use of equity financing and also reduce their debt as the cost of issuing equity increases. That is, these firms are unable to substitute equity financing for debt financing when external equity financing is too costly.

Quantitatively, the model matches not only cross-sectional moments of asset prices, but also key properties of the firm-level investment rate, and debt and equity financing (among other moments). Through several comparative static exercises, we show that the existence of positive and time-varying external equity issuance costs is important for the good quantitative fit of the model. Without external equity financing costs, the model generates investment rates that are too volatile (0.34 compared with 0.22 in the baseline model and the data) and value and investment return spreads that are too small. These results are intuitive. Without external equity financing costs, all firms take advantage of this cost-free marginal source of financing to smooth their payouts in response to the shocks, thus significantly reducing the endogenous risk dispersion in the cross-section. In addition, when external equity financing is costly but time

invariant, the model implies a counterfactually too high correlation of equity issuance-to-asset ratio and sales growth (25% vs. 11% in the data) and very small cross-sectional risk premiums. On the debt financing margin, when we significantly tighten the collateral constraint or increase the debt adjustment cost, all the cross-sectional return spreads become significantly smaller. This result is also intuitive. Under these specifications, all firms effectively face a large cost of substituting debt for equity and hence tend to save in cash due to a precautionary saving motive. As a result, both high and low productive firms use cash to smooth negative aggregate shocks which substantially reduces the endogenous risk dispersion in the cross-section.

1. Related Literature

This paper is related to the literature that examines the impact of financial frictions on corporate investment and asset prices.¹ The work most closely related to ours is Bolton, Chen, and Wang (2013), who study firms' investment, financing, and cash management decisions in a dynamic q -theoretic framework in which, similar to our model, external financing conditions are stochastic, and Eisfeldt and Muir (2016), who infer the aggregate cost of external (debt and equity) finance by using firms' cross-sectional investment, financing, and saving decisions in a dynamic model. Indeed, the use of firm-level cross-sectional data to construct an empirical proxy of the aggregate shock to the cost of equity issuance closely follows the empirical approach in Eisfeldt and Muir (2016). Our analysis is complementary to these studies in that we focus on the impact of the time varying external equity issuance cost on risk premiums in the cross-section, a dimension that is not examined in these studies. Additionally, Lamont, Polk, and Saa-Requejo (2001) find that constrained firms are subject to common shocks but have low average returns in a sample of manufacturing firms from 1968 to 1997. We differ in that we explore both theoretically and empirically the implications of equity ICS for asset prices and corporate financing policies.

The micro-foundations of time-varying equity issuance costs, and its impact on investor's marginal utility (stochastic discount factor), include endogenous time-varying adverse selection problems in Eisfeldt (2004), Kurlat (2013), and Bigio (2015), agency frictions varying over time like in Bernanke and Gertler (1989) and Carlstrom and Fuerst (1997), time-varying endogenous information production like in Dow, Goldstein, and Guembel (2017), changes in investors' sentiment like in Baker and Wurgler (2002, 2006), time-varying aggregate liquidity like in Pastor and Stambaugh (2003), time-varying risk aversion of financial intermediaries like in Adrian, Moench, and Shin (2012), Altinkilic and Hansen (2000) on investors' willingness to supply equity capital to firms.

¹ See, for example, Hennessy and Whited (2005, 2007), Carlson, Fisher, and Giammarino (2006), Bolton, Chen, and Wang (2011), DeMarzo et al. (2012), and Bolton, Wang, and Yang (2014), among others.

To the extent that these frictions affect equilibrium allocations, they are likely to affect investor's equilibrium marginal utility as well. The variation in aggregate equity issuance costs may also affect investor's marginal utility because a large portion of research and development (R&D) expenditures, an important driver of long-run growth prospects (hence, of long-run risks), is equity financed, like in Bianchi, Kung and Morales (2018). We do not explain the sources of the variation in equity issuance costs nor its impact on investor's marginal utility. Instead, we take the evidence from these previous studies and incorporate these costs in a dynamic model of the firm with financial frictions to understand the impact of time variation in equity issuance costs on asset prices in the cross-section.

The fact that equity issuance is costly for firms is well established in the corporate finance literature. The equity issuance costs include both direct costs (e.g., flotation costs: underwriting, legal, and registration fees), and indirect (unobserved) costs due to asymmetric information, managerial incentive problems, among others as discussed previously. These costs can be substantial. For example, Altinkilic and Hansen (2000) estimate the underwriting fee ranging from 4.37% to 6.32% of the capital raised in their sample. In addition, a few empirical papers also seek to establish the importance of the indirect costs of equity issuance. Asquith and Mullins (1986) find that the announcement of equity offerings reduces stock prices on average by -3% and this price reduction as a fraction of the new equity issue is on average -31% .

Furthermore, the cost of issuing equity varies over time. Choe, Masulis, and Nanda (1993) find that the adverse selection costs measured as negative price reaction to seasoned equity offerings (SEO) announcement is higher in contractions and lower in expansions. Erel et al. (2012) show that firms' access to external finance markets also changes with macroeconomic conditions. Kahle and Stulz (2013) find that net equity issuance falls more substantially than debt issuance during the recent financial crisis suggesting that shocks to the corporate credit supply are not likely to be the cause for the reduction in firms' capital expenditures in 2007-2008. McLean and Zhao (2014) document that both investment and employment are more sensitive to cash flows during recessions. Consistent with our results, McLean and Zhao (2014) also find that equity issuance plays a bigger role than debt issuance in causing their finding. Motivated by the findings in Covas and Den Haan (2011), Begenau and Salomao (2018) study the differential financing behavior of small and big firms over the business cycles. We contribute to this literature by proposing a novel proxy for the aggregate shock to the cost of external equity issuance, which we use to show that this shock is an important determinant of firm's equilibrium risk premium.

A related literature that studies asset prices in production economies has primarily focused on aggregate shocks that originate in the real sector, for example, aggregate productivity shocks, investment-specific shocks (e.g., Kogan and Papanikolaou 2013, 2014) and adjustment cost shocks

(e.g., Belo, Lin, and Bazdresch 2014), or shocks that originate from changes in monetary and fiscal policies.² Our paper differs in that we focus on financial shocks which allow us to provide a novel mechanism to explain the empirical failure of the standard capital asset pricing model (CAPM) that is different from the existing investment-based literature that highlights the role of either investment shocks or adjustment cost shocks. This mechanism is also different from Kojien, Lustig, and Van Nieuwerburgh (2017), who highlight the different exposure of value and growth firms to shocks signaling future economic growth, and Garleanu, Kogan, and Panageas (2011), who study the link between displacement risk and asset returns. In addition, our approach of extracting the common risk factor (equity ICS) from firms' equity issuances is similar in spirit to Herskovic et al. (2016), who show that the common factor in idiosyncratic volatility is a priced factor in the cross-section of stock returns.

Recent work in macroeconomics investigates the impact of financial shocks (frictions) on aggregate quantities. Different from the shocks to the credit supply highlighted in Jermann and Quadrini (2012) and Khan and Thomas (2013), we focus on an aggregate shock to the cost of external equity issuance and on the implications of this shock for risk premiums in the cross-section. The financial frictions in our model are similar in spirit to Bernanke and Gertler (1989), Kiyotaki and Moore (1997), and Bernanke, Gertler, and Gilchrist (1999), among others. The difference is that disturbance in the financial sector acts as a source of aggregate economic fluctuations in our model (like in Jermann and Quadrini 2012; Khan and Thomas 2013) as opposed to propagating shocks that originate in other sectors of the economy.

2. Model

In this section, we present a dynamic model economy with time-varying external equity financing frictions to investigate the impact of equity ICS on asset prices and financial flows in the cross-section.

2.1 Technology

A large number of firms produce a homogeneous good. Firms use physical capital (K_t) to produce this good (Y_t). To save on notation, we omit firm index j whenever possible. The production function is given by

$$Y_t = Z_t X_t^{1-\theta} K_t^\theta, \quad (1)$$

in which X_t is aggregate productivity and Z_t is firm-specific productivity. The production function exhibits decreasing returns to scale, that is, $0 < \theta < 1$.

² For example, Jermann (1998), Kaltenbrunner and Lochstoer (2010), and Favilukis and Lin (2016), among others, study the asset pricing implications of aggregate productivity shocks. Papanikolaou (2011) and Kogan and Papanikolaou (2013, 2014) focuses on investment-specific shocks. Gilchrist and Leahy (2002) study the relationship between monetary policy and asset prices. Kung et al. (2012) explore the market price of fiscal policy risk.

Aggregate productivity follows a random walk process with a drift

$$\Delta x_{t+1} = \mu_x + \sigma_x \varepsilon_{t+1}^x, \quad (2)$$

in which $x_{t+1} = \log(X_{t+1})$, Δ is the first-difference operator, ε_{t+1}^x is an i.i.d. standard normal shock, and μ_x and σ_x are the average growth rate and conditional volatility of aggregate productivity, respectively.

Firm-specific productivity follows the AR(1) process

$$z_{t+1} = \bar{z}(1 - \rho_z) + \rho_z z_t + \sigma_z \varepsilon_{t+1}^z, \quad (3)$$

in which $z_{t+1} = \log(Z_{t+1})$, ε_{t+1}^z is an i.i.d. standard normal shock that is uncorrelated across all firms in the economy and independent of ε_{t+1}^x , and $\bar{z}$, ρ_z , and σ_z are the mean, autocorrelation, and conditional volatility of firm-specific productivity, respectively.

Physical capital accumulation is given by

$$K_{t+1} = (1 - \delta)K_t + I_t, \quad (4)$$

where I_t represents investment and δ denotes the capital depreciation rate.

We assume that capital investment entails convex asymmetric adjustment costs, denoted by G_t , which are given by

$$G_t = \begin{cases} \frac{c_k^+}{2} \left(\frac{I_t}{K_t} \right)^2 K_t, & I_t \geq 0 \\ \frac{c_k^-}{2} \left(\frac{I_t}{K_t} \right)^2 K_t, & I_t < 0, \end{cases} \quad (5)$$

where c_k^+ and c_k^- determine the upward and downward speed of adjustment, respectively. We allow capital adjustment costs to be asymmetric to capture costly reversibility of capital. The costly reversibility can arise because of resale losses due to transaction costs or the market for lemons phenomenon.

2.2 Collateral constraints and retained earnings

Firms use equity and debt to finance investment. At the beginning of time t , firms can issue an amount of debt, denoted by B_t , which must be repaid at the beginning of period $t + 1$. Firms can also save using cash, in which case B_t takes on negative values.³ The firm's ability to borrow is bounded by the limited enforceability as firms could default on their obligations. Following Hennessy and Whited (2005), we assume that the only asset available for liquidation is the physical capital stock K_{t+1} . In particular, we require the liquidation value

³ We treat cash as negative debt for tractability, given the already high-dimensional dynamic problem. In principle, negative debt and cash may not be perfect substitutes if their marginal costs are different, like in Acharya, Almeida, and Campello (2007) and Bolton, Chen, and Wang (2014).

of capital to be greater than or equal to the debt payment. It follows that the collateral constraint is given by

$$B_{t+1} \leq \varphi K_{t+1}. \quad (6)$$

The variable $0 < \varphi < 1$ affects the tightness of the collateral constraint, and therefore, the borrowing capacity of the firm. Because of the collateral constraint, the interest rate, denoted by r_f , is the risk-free rate. This risk-free rate is constant, consistent with the specification of the stochastic discount factor which we discuss in Section 2.4.

The interest rate on corporate savings, r_s , is lower than the borrowing risk-free rate, r_f . This is because holding cash is costly for firms (Graham 2000). If the two rates are equal, firms save all free cash flows and do not distribute cash to shareholders in the presence of external financing frictions. Following Hennessy, Levy, and Whited (2007) and Livdan, Saprizza, and Zhang (2009), we assume the saving rate is smaller than the borrowing rate so that firms are not indifferent between savings and cash distributions. That is,

$$r_s = r_f - \kappa, \quad (7)$$

where $\kappa > 0$ captures the wedge between the borrowing and saving rates. Let

$$r_l = r_f \mathbf{1}_{\{B_t > 0\}} + r_s \mathbf{1}_{\{B_t < 0\}} \quad (8)$$

denote the applicable interest rate to the firm.

Firms also incur adjustment costs, denoted by Φ_t , when changing the amount of debt outstanding,

$$\Phi_t = \frac{c_b}{2} \left(\frac{\Delta B_{t+1}}{B_t} \right)^2 B_t \mathbf{1}_{\{B_t > 0\}}, \quad (9)$$

where $\Delta B_{t+1} = B_{t+1} - B_t$. Firms can freely adjust their cash holdings, so $\Phi_t = 0$ if $B_t < 0$. Debt adjustment costs capture the fact that adjusting capital structure is costly. The convexity in the adjustment cost function generates a more persistent debt growth process, and hence allows us to generate a persistent leverage process, consistent with the empirical evidence (Leary and Roberts 2005). We note that the debt adjustment cost is calibrated to be very low in the baseline model (as we discuss in the calibration Section 2.6).

2.3 Costly external equity financing

Taxable corporate profits are equal to output less capital depreciation and interest expenses: $Y_t - \delta K_t - r_f B_t \mathbf{1}_{\{B_t \geq 0\}}$. It follows that the firm's budget constraint can be written as

$$E_t = (1 - \tau)Y_t + \tau \delta K_t + \tau r_f B_t \mathbf{1}_{\{B_t \geq 0\}} - I_t - G_t + B_{t+1} - (1 + r_l)B_t - \Phi_t, \quad (10)$$

in which τ is the corporate tax rate, $\tau \delta K_t$ is the depreciation tax shield, $\tau r_f B_t \mathbf{1}_{\{B_t \geq 0\}}$ is the interest tax shield, and E_t is the firm's payout.

When the sum of investment, capital, and debt adjustment costs exceeds the sum of after tax operating profits and debt financing, firms can take external funds by means of seasoned equity offerings. External equity H_t is given by

$$H_t = \max(-E_t, 0). \quad (11)$$

As discussed in the Introduction section, issuing equity is costly for firms and this cost may vary over time. Like in Bolton, Chen, and Wang (2013), we do not explicitly model the sources of this cost. Rather, we attempt to capture the effect of this cost in a reduced-form fashion. We assume that the external equity cost includes a fixed, a linear, and a quadratic component, following Hennessy and Whited (2007), and is stochastic, like in Bolton, Chen, and Wang (2013). The assumption of linear and convex issuance cost is also consistent with Myers and Majluf (1984) and Krasker (1986), who show that the (marginal) cost of external equity is increasing in asymmetric information in equity markets. Specifically, we parameterize the equity issuance cost function as⁴

$$\Psi(H_t) = \left(\eta_0 X_t + \eta_1 H_t + \eta_2 \frac{H_t^2}{K_t} \right) \exp[-\eta_3 \xi_t] \mathbf{1}_{\{H_t > 0\}}, \quad (12)$$

in which ξ_t captures the time-varying cost of external equity financing. This stochastic cost follows an AR(1) process,

$$\xi_{t+1} = \rho_\xi \xi_t + \sigma_\xi \varepsilon_{t+1}^\xi, \quad (13)$$

in which ρ_ξ and σ_ξ are the first-order autocorrelation coefficient and conditional volatility of ξ_{t+1} , and ε_{t+1}^ξ is an i.i.d. standard normal shock that is independent of ε_{t+1}^x and ε_{t+1}^z . In Equation (12), periods in which ξ_t is high are associated with low costs of external equity financing, and vice versa.

Finally, firms do not incur costs when paying dividends. The effective cash flow D_t distributed to shareholders is given by

$$D_t = E_t - \Psi_t. \quad (14)$$

2.4 Firm's problem

The firm takes as given the stochastic discount factor $M_{t,t+1}$ used to value the cash flows arriving in period $t+1$ (and subsequent periods). We specify the stochastic discount factor to be a function of the two aggregate shocks in the economy:

$$M_{t,t+1} = \frac{1}{1+r_f} \frac{e^{-\gamma_x \Delta x_{t+1} - \gamma_\xi \Delta \xi_{t+1}}}{\mathbb{E}_t \left[e^{-\gamma_x \Delta x_{t+1} - \gamma_\xi \Delta \xi_{t+1}} \right]}, \quad (15)$$

where r_f is the risk-free rate. The sign of the risk factor loading parameters (γ_x and γ_ξ) is positive, consistent with the evidence reported in the empirical

⁴ Note that aggregate productivity, X_t , is included for the fixed cost component to ensure the economy is cointegrated along the balanced growth path; this term cancels out in the stationary representation of the model.

section below. The risk-free rate is set to be constant. This allows us to focus on risk premiums as the main driver of the results in the model as well as to avoid parameter proliferation.

Firms choose capital investment and debt/cash optimally by solving the following maximization problem:

$$V_t = \max_{I_t, K_{t+1}, B_{t+1}} D_t + \mathbb{E}_t[M_{t,t+1} V_{t+1}], \quad (16)$$

subject to firms' capital accumulation Equation (4), collateral constraint Equation (6), budget constraint Equation (10), and cash-flow Equation (14).

2.5 Equilibrium risk and return

In the model, risk and expected stock returns are determined endogenously along with the firm's optimal investment and financing decisions. To make the link explicit, we evaluate the value function in Equation (16) at the optimum and obtain

$$V_t = D_t + \mathbb{E}_t[M_{t,t+1} V_{t+1}], \quad (17)$$

$$\Rightarrow 1 = \mathbb{E}_t[M_{t,t+1} R_{t+1}^s], \quad (18)$$

in which Equation (17) is the Bellman equation for the value function, and the Euler Equation (18) follows from the standard formula for stock return $R_{t+1}^s = V_{t+1}/[V_t - D_t]$. Substituting the stochastic discount from Equation (15) into Equation (18), and some algebra, yields the following equilibrium pricing equation:⁵

$$\mathbb{E}_t[r_{t+1}^e] = \gamma_x \times \text{Cov}(r_{t+1}^e, \Delta x_{t+1}) + \gamma_\xi \times \text{Cov}(r_{t+1}^e, \Delta \xi_{t+1}), \quad (19)$$

in which $r_{t+1}^e = R_{t+1}^s - R_f$ is the stock excess return, and $R_f \equiv 1 + r_f = \mathbb{E}_t[M_{t,t+1}]^{-1}$ is the gross risk-free rate.

According to Equation (19), equilibrium risk premiums in the model are determined by the endogenous covariances of firms' excess stock return with the two aggregate shocks (quantity of risk) and by the stochastic discount factor risk factor loadings on the two risk factors in Equation (15) (γ_x and γ_ξ). The assumption of a positive sign of the risk factor loadings (and hence positive prices of risk) implies that, all else equal, assets with returns that have a high positive covariance with the aggregate productivity shock are risky, and hence offer high average returns in equilibrium. Similarly, all else equal, assets with returns that have a high positive covariance with the aggregate equity issuance cost shock are risky, and hence offer high average returns in equilibrium.

⁵ This derivation is standard. Equation (18) implies $\mathbb{E}_t[M_{t,t+1}(R_{t+1}^s - R_f)] = 0$, because $\mathbb{E}_t[M_{t,t+1}]R_f = 1$. Using a first-order log-linear approximation of the stochastic discount factor (SDF) $M_{t,t+1}$ defined in Equation (15), and applying the formula for covariance $\text{Cov}(X, Y) = \mathbb{E}[XY] - \mathbb{E}[X]\mathbb{E}[Y]$ to the previous equation, plus some algebra, yields Equation (19).

Table 1
Calibration

Parameter	Symbol	Value
<i>Technology</i>		
Returns to scale	θ	0.7
Corporate tax rate	τ	0.2
Rate of depreciation for capital	δ	0.01
Adjustment cost parameters in capital	c_k^+/c_k^-	0/40
Adjustment cost parameters in debt	c_b	0.3
Resale value of capital	φ	0.85
Wedge between the borrowing and saving rates	κ	0.005/12
Fixed/linear/convex equity issuance costs	$\eta_0/\eta_1/\eta_2$	0.002/0.1/0.0004
Parameter of time-varying equity issuance cost	η_3	12
<i>Stochastic processes</i>		
Average growth rate of aggregate productivity	μ_x	0.013/12
Conditional volatility of aggregate productivity	σ_x	0.055
Average level of firm-specific productivity	$\bar{z}$	-3.9
Persistence of firm-specific productivity	ρ_z	0.98
Conditional volatility of firm-specific productivity	σ_z	0.15
Persistence of issuance disturbance	ρ_ξ	0.98
Conditional volatility of issuance disturbance	σ_ξ	0.035
Loading of the SDF on the aggregate TFP shock	γ_x	6.5
Loading of the SDF on the equity ICS	γ_ξ	12
Risk-free rate	r_f	0.0165/12

This table presents the calibrated parameter values of the baseline model.

2.6 Calibration

All of the endogenous variables in the model are functions of the state variables. Because the functional forms are not available analytically, we solve for these functions numerically. Appendix A-2 provides a description of the solution algorithm (value function iteration) and the numerical implementation of the model.

The model is solved at a monthly frequency. Table 1 reports the parameter values used in the baseline calibration. We use parameter values reported in previous studies, whenever possible, or by matching the selected moments in the data reported in Table 2.⁶ To evaluate the model fit, the table reports the target moments in both the data and the model. To generate the model's implied moments, we simulate 3,600 firms for 1,000 monthly periods. We drop the first 400 months to neutralize the impact of the initial condition. The remaining 600 months of simulated data are treated as those from the economy's stationary distribution. We then simulate 100 artificial samples and report the cross-sample average results as model moments.

2.6.1 Firm's technology: General parameters. We set the curvature of the production function θ to be 0.7, close to the value estimated by Cooper and Ejarque (2001) and Hennessy and Whited (2007). The capital depreciation rate δ is set to be 1% per month, like in Bloom (2009). We set the corporate tax rate

⁶ Because all the firm-level accounting variables in the data are only available at an annual frequency, we time aggregate the simulated accounting data to make the model-implied moments comparable with those in the data.

Table 2
Selected moments

Moments	Data	Model
<i>Asset prices</i>		
Aggregate stock market excess return (%)	5.71	5.58
Sharpe ratio aggregate stock market	0.35	0.32
Real risk-free rate (%)	1.65	1.65
<i>Real quantities: Aggregate level</i>		
SD of profits	0.14	0.15
SD of debt growth rate	0.09	0.10
SD of model-implied equity ICS	0.102	0.107
Average fraction of equity issuance	0.17	0.21
Marginal equity issuance cost	.084–.12	0.10
<i>Real quantities: Cross-section</i>		
SD of investment rate	0.25	0.22
SD of equity issuance-to-asset ratio	0.13	0.08
Autocorrelation of investment rate	0.41	0.33
Autocorrelation of equity issuance-to-asset ratio	0.30	0.27
Interdecile range of investment rate	0.57	0.45
Interdecile range of equity issuance-to-asset ratio	0.10	0.13
Skewness of investment rate	1.26	2.18
Skewness of equity issuance-to-asset ratio	4.55	3.08
Mean of equity issuance-to-asset ratio	0.04	0.05
Mean of leverage ratio	0.25	0.23
Autocorrelation of leverage	0.86	0.84

This table presents the selected target moments used for the calibration of the baseline model. We compare the moments in the data with moments of simulated data. The model-implied moments are the mean value of the corresponding moments across simulations. The real data are from 1964 to 2013. The data moment for marginal equity issuance cost is from Hennessy and Whited (2007). The reported statistics for the model are obtained from 100 samples of simulated data, each with 3,600 firms and 600 monthly observations.

to be 0.20 consistent with Gomes and Schmid (2010). We set the liquidation value parameter $\varphi = 0.85$ and the wedge between borrowing and saving rate κ to be 0.005/12, following Livdan, Saprizza, and Zhang (2009).

2.6.2 Firm's technology: Equity issuance costs and capital/debt adjustment costs. We calibrate the parameters of the equity issuance cost function $\{\eta_0, \eta_1, \eta_2, \eta_3\}$ to be $\{0.002, 0.1, 0.0004, 12\}$, by matching the fraction of equity issuance, the mean, and autocorrelation and interdecile range of the firm's gross equity issuance-to-asset ratio. In the model, these moments are 0.21, 0.05, 0.27, and 0.13, close to the data moments of 0.17, 0.04, 0.30, and 0.10, respectively. Additionally, the model-implied volatility and skewness of equity issuance-to-asset ratio (not targeted) are 0.08 and 3.1, respectively, close to the data moments of 0.13 and 4.6, respectively.

Furthermore, the model-implied equity issuance costs have reasonable economic magnitudes, and are consistent with the evidence from the existing literature. Because indirect equity costs are not observable, we refer to the data estimate of direct issuance costs from Altinkilic and Hansen (2000), which serve as a lower bound estimate of equity issuance costs. The model-implied total equity issuance costs including both direct and indirect costs is 13% of the issuance proceeds; not surprisingly, it is bigger than the direct costs estimates of 5.4% in the data. Thus our finding implies that the indirect issuance costs are large, consistent with Hennessy and Whited (2007), who find large indirect

issuance costs in a structural model (without variation in the cost of equity issuance). Specifically, our marginal costs of equity issuance is 10%, well within with the range of estimates in Hennessy and Whited (2007).

Given the calibrated equity issuance costs parameters, we calibrate the capital and debt adjustment cost parameters to match several cross-sectional and time-series moments of firms' investment rates and debt growth rates. We set the convex capital adjustment costs to be $c_k^+ = 0$, and $c_k^- = 40$. Table 2 shows that this calibration of the model matches reasonably well the volatility, autocorrelation, interdecile range, and skewness of firm-level investment rates. We set the debt adjustment cost to be $c_b = 0.3$ to match the cash-to-asset ratio at 0.05 (conditional on firms holding cash). This parameter also implies that the volatility of the aggregate debt growth rate is 0.10 close to the data moment at 0.09. The implied mean and autocorrelation of the firm-level financial leverage ratio are 0.23 and 0.84, respectively, close to the data moments at 0.25 and 0.86. This calibration generates fairly small capital adjustment costs, 0.2% of the firm sales, at the lower end of the empirical estimates in the literature (Hamermesh and Pfann 1996; Bloom 2009). It also generates very small debt adjustment costs, about 0.9% of the firm sales.

2.6.3 Stochastic processes. In the model, the aggregate productivity shock is essentially a profitability shock. We set the conditional volatility of the aggregate productivity shock to be $\sigma_x = 0.055$ to match the volatility of aggregate profits (0.14 in the data and 0.15 the model). In the data, we measure aggregate profits using data from the National Income and Product Accounts (NIPA). We set $\mu_x = .013/12$, which implies the average growth rate of aggregate productivity is 0.013 at annual frequency, consistent with the data. Given the volatility of the aggregate productivity shock, we set the conditional volatility of the aggregate equity ICS to be $\sigma_\xi = 0.035$ and its persistence to be $\rho_\xi = 0.98$, so that the volatility of model-implied equity ICS is close to the data (10.7% vs. 10.2% in the data). In addition, the implied firm-level correlation of gross equity-issuance-to-book-equity ratio and sales growth is 0.11, consistent with the data at 0.11 as well. Through experimentation, we find that without a volatile equity ICS, the implied correlation between equity issuance and sales growth is counterfactually too high.

We calibrate the persistence and conditional volatility of the firm-specific productivity shock to be $\rho_z = 0.98$ and $\sigma_z = 0.15$, respectively, which implies a firm level return volatility of 0.33, consistent with Campbell et al. (2001). The long-run average level of firm-specific productivity, $\bar{z}$, is a scaling variable. We set $\bar{z} = -3.9$, which implies that the long run average debt in the economy is 1. To calibrate the stochastic discount factor, we set the real risk-free rate to be $r_f = 1.65\%$ per annum. Given the calibrated conditional volatility of aggregate shocks, we set the stochastic discount factor risk factor loadings on the aggregate productivity shock and aggregate equity ICS to be $\gamma_x = 6.5$ and $\gamma_\xi = 12$, respectively, by matching the average aggregate stock market return

and its Sharpe ratio as close as possible. This calibration generates an average market excess return of 5.6% and Sharpe ratio of 0.32, reasonably close to 5.7% and 0.35, respectively, in the data.

3. Estimation of Equity Issuance Cost Shocks

As discussed in Section 1, the firm's total cost of issuing equity includes both direct and indirect (which are unobserved) costs. Because data on these total costs are not readily available, we use the structure of the model to estimate an aggregate model-implied equity ICS proxy in the real data for the U.S. economy from 1964 to 2013. The estimation of the equity ICS proxy is important in the context of our analysis given its central role in the model, thus allowing us to test the models' assumptions and economic mechanisms in the real data.

3.1 A model-implied equity ICS proxy

Following Eisfeldt and Muir (2016), we identify through a multivariate linear regression a set of moments in the model that explain most of the variation of the model's equity ICS ($\Delta\xi$). We find that the following two moments, the change in the fraction of firms issuing equity in the cross-section (denoted by ΔEI), and the change in the aggregate debt-to-physical capital ratio (denoted by ΔDK), when combined, capture almost all of the variation of the equity ICS in the model. Intuitively, the first moment captures changes in firms' extensive margin of external equity issuance when the cost of issuing equity varies, while the second moment captures the substitution effect between debt and equity when the relative cost of these sources of external financing changes. We then use the regression slope coefficients on these two moments obtained from the model estimates, along with the corresponding moments at each date in the real data, to construct a time series of the equity ICS proxy.

Specifically, we run the following regression in simulated data generated from the baseline calibration of the model:

$$\Delta\xi_t = \beta_1 \Delta EI_t + \beta_2 \Delta DK_t + \epsilon_t \quad (20)$$

where ΔEI and ΔDK are normalized to have mean zero and unit standard deviation. To be consistent with the frequency of the real data, we estimate the regression using data from the model aggregated at the annual frequency. Table 3 reports the estimation results. The slope coefficients are $\beta_1 = 0.046$ and $\beta_2 = -0.078$, both of which are statistically significant. Intuitively, the slope coefficient on the change in the fraction of firms issuing equity in the cross-section is positive because when $\Delta\xi$ is high, the cost of issuing equity is low, which leads to an increase in the number of firms issuing equity. The negative sign on ΔDK captures the fact that firms tend to issue more debt to substitute for equity when the equity issuance cost shock $\Delta\xi$ is low. Most important, the R^2 of the regression is 86%, which implies that these two moments capture most of the variation in the true equity ICS in the model. Because the equity

Table 3
Estimation of an aggregate equity issuance cost shock proxy

	(1)	(2)
ΔEI	0.090	0.046
$[t]$	26.44	21.17
ΔDK		-0.078
$[t]$		-31.55
R^2	.56	.86

This table reports the estimation results from two alternative specifications of the following linear regression:

$$\Delta \xi_t = \beta_1 \Delta EI_t + \beta_2 \Delta DK_t + \epsilon_t,$$

in which $\Delta \xi_t$ is the exogenous equity issuance cost shock in the model, ΔEI is the standardized change in the fraction of firms issuing equity in the cross-section, and ΔDK is the standardized change in the aggregate debt-to-capital ratio. $[t]$ is the corresponding slope t -statistic, and R^2 is the regression R^2 . Specification 1 includes ΔEI only, and specification 2 includes both ΔEI and ΔDK . The estimation is based on simulated data from the baseline calibration of the model and is performed at the annual frequency.

ICS proxy is constructed using the fitted values of Equation (20), this regression R^2 implies that the correlation between the true equity ICS and its proxy in the model is quite high, 93%, which provides support for the use of this equity ICS proxy in the real data. As robustness checks, we have also investigated adding more moments on the right-hand side of Equation (20), for example, the cross-sectional correlation between debt growth and equity issuance, the fraction of positive debt growth, among other moments, but found the impact of adding these moments on the regression R^2 to be fairly small.

We then apply the model-implied regression slope coefficients β_1 and β_2 to the analogous empirical moments in the data to construct a time series of the equity ICS proxy from 1964 to 2013 in the U.S. economy. We compute the empirical moments using data for publicly traded firms in Compustat Annual files. ΔEI is given by the change in the percentage of firms in the cross-section that issue equity. We classify a firm as an equity issuer in a given year if its gross equity issuance (data item SSTK, sale of common and preferred stock) in that year is greater than or equal to 3% of its average year-begin and year-end market equity.⁷ We impose this threshold to focus on the active (management-initiated) issuances, which is the empirical counterpart of equity issuance in the model, as opposed to passive (employee-initiated) issuances, which are outside the scope of the model. The Compustat variable SSTK contains the exercise of stock options to employees as a form of compensation (among other forms) and, as documented in Fama and French (2005) and McKeon (2015), these passive issuances are large relative to SEOs and private placements. Using several data

⁷ Compustat data item SSTK is only available after 1971. We extend the fraction of equity issuing firms back to 1964 using a measure of net equity issuance from Fama and French (2008). In particular, we define net equity issuance as the change in log split-adjusted shares. The split-adjusted shares are the products of common shares outstanding (annual Compustat data item CSHO) and the stock split and dividend adjustment factor (annual Compustat data item AJEX). Similarly, we classify a firm as an issuing firm in a year if its log change in shares is greater than or equal to 3% to exclude small issuances due to the exercising of employees stock options. It is valid to use net equity issuance to approximate gross equity issuance in 1960s, because equity repurchases were rare before 1980 (see Grullon and Michaely 2002; Boudoukh et al. 2007).

sources, McKeon (2015) shows that using the 3% threshold on market value purges the vast majority of passive equity issuances from the Compustat data, without sacrificing a substantial number of the active issuances.

ΔDK is given by the change in the aggregate debt-to-physical capital ratio (debt is the sum of data items DLTT and DLC, short term and long term debt, and physical capital is data item PPENT, net property, plant and equipment. Both variables are aggregated across firms). In computing this aggregate ratio, we exclude the top 1% largest firms measured by market equity to mitigate the impact of the very large firms on this measure. This approach is motivated by the findings in Covas and Den Haan (2011), who show that external finance for the largest firms (especially those at the top 1% of the size distribution) is not representative of the financing behavior of the rest of the firms in the economy because their issuance is either acyclical or countercyclical, in contrast with the behavior of almost all of the other firms in Compustat, for which debt and equity issuances are procyclical. The Internet Appendix shows that our main results are robust to reasonable alternative ways of constructing the variables ΔEI and ΔDK .

3.2 Properties of the equity ICS proxy

Panel A in Figure 1 shows the time series of the equity ICS proxy, and panel B shows the time series of ΔEI and ΔDK , the two variables used to construct the equity ICS proxy. The figure shows that the equity ICS proxy tends to be low in the early stage of recessions indicating high cost of equity issuance in these periods. From panel B, we can see that the low equity ICS in these years is due to both a low fraction of firms issuing equity and high aggregate debt-to-capital ratio (the regression slope on aggregate debt-to-capital ratio in Equation (20) is negative) in the early stage of recessions. Finally, panel A shows that, despite some correlation with the business cycle, the equity ICS proxy captures variation in economic conditions that transcends the NBER-designated business-cycle recessions dates. For example, equity issuance costs increased substantially (low equity ICS realizations) in 1985 and 1988, but these years are not classified by the NBER as recession years. This evidence suggest that financial conditions are not perfectly correlated with the business cycle, providing support for our modeling approach in which aggregate economic conditions are driven by two (financial and productivity) aggregate shocks, not just one (productivity shocks, the driver of the business cycle in standard RBC models).

Panel A in Table 4 reports the descriptive statistics of the equity ICS proxy, changes in aggregate total factor productivity (ΔTFP), and additional variables. Panel B reports the correlation of the equity ICS proxy with several macroeconomic variables.

3.2.1 Summary statistics. As reported in panel A in Table 4, the equity ICS proxy and the change in TFP (which are the two aggregate shocks in the model),

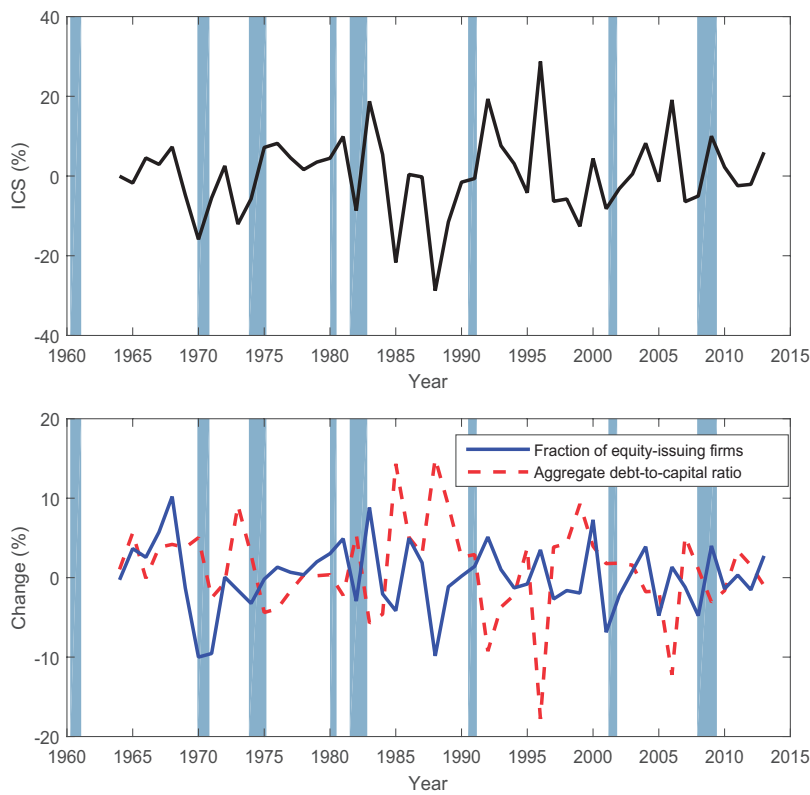


Figure 1
Model-implied equity issuance cost shock proxy
This figure reports the time series of the model-implied equity ICS proxy (top panel) and the time series of the change in the fraction of equity issuance and the ratio between aggregate debt to aggregate capital (bottom panel). Shaded bars represent NBER recession years. Data are annual from 1964 to 2013.

have very different volatilities in the data. The equity ICS proxy is significantly more volatile than the changes in aggregate total factor productivity, with a standard deviation of 10.2% per year and 1.0% per year, respectively. This evidence suggest that external equity financing conditions vary substantially over time.

3.2.2 Relationship with time-varying investment opportunities and cost of debt. One possible concern with the interpretation that the equity ICS proxy in the empirical analysis captures time-variation in the aggregate cost of equity issuance, is that this proxy relies on a model that is a simplified version of the real economy, and hence it might be contaminated by other aggregate shocks in the data. For example, the equity ICS proxy measure in the real data might be contaminated by time-varying investment opportunities, if investment opportunities vary more in the real data than in the model, or by time-varying

Table 4
Properties of equity issuance cost shock proxy

A. Summary statistics

Variable	Mean	SD	AR(1)
Fraction of firms issuing equity	17.02	5.77	0.74
Change in fraction issuing equity (ΔEI)	0.00	4.20	0.05
Change in aggregate debt-to-capital ratio (ΔDK)	1.25	5.98	0.08
Change in total factor productivity (ΔTFP)	1.13	1.34	0.12
Model-implied equity issuance cost shock (ICS)	0.00	10.23	0.01

B. Correlations

	Group 1					Group 2			
	ΔTFP	CAPE	$\Delta CAPE$	ISTS	ΔC	ΔJQ	ΔEM	$\Delta B-A$	RF
ICS	-0.02	-0.03	0.17	-0.08	-0.02	-0.16	-0.18	-0.16	-0.07
<i>p</i> -value	.91	.82	.23	.56	.90	.44	.33	.26	.62

Panel A reports the summary statistics (mean, standard deviation, denoted by SD, and first-order autocorrelation, denoted by AR(1)) of the time series of the fraction of firms issuing equity in the cross-section, the change in fraction of firms issuing equity in the cross-section (ΔEI), the change in aggregate debt-to-capital ratio (ΔDK), the change in aggregate total factor productivity (ΔTFP), and the equity issuance cost shock proxy (ICS) in the data. Both mean and standard deviation are in percentage terms. Panel B reports the correlation of the model-implied equity ICS proxy with several aggregate shocks and macroeconomic variables. In Group 1, we consider the following proxies of time-varying investment opportunities: ΔTFP ; the cyclically adjusted price-to-earnings ratio (CAPE) and its change ($\Delta CAPE$); a proxy of investment-specific technological shocks (ISTS), measured as the real quality-adjusted investment price growth; and the growth rate of per capita real nondurables consumption (ΔC). In Group 2, we consider the following financial shock variables: the Jermann and Quadrini (2012) collateral-constraint shock (ΔJQ); an external (equity plus debt) finance shock from Eisfeldt and Muir (2016) (ΔEM); the change in the Moody's BBB-AAA credit spread ($\Delta B-A$); and the risk-free rate (RF). Benchmark data are annual from 1964 to 2013, but correlations with some variables are based on shorter samples because of data availability.

cost of debt, which is absent from the model. Indeed, in the real economy, it is natural to expect that variation in investment opportunities and cost of debt also drive equity issuance and leverage decisions, the variables used to construct the equity ICS proxy.

To address this concern, we examine the correlation between the equity ICS proxy and other aggregate macroeconomic variables related to time-varying investment opportunities and cost of debt (broadly defined). As measures of investment opportunities (Group 1 in panel B) in the economy, we consider changes in aggregate TFP, the level and growth rate of the Shiller's cyclically adjusted price-to-earnings ratio (CAPE and $\Delta CAPE$, respectively), a proxy of investment-specific technology shocks (ISTS, measured as the real quality-adjusted investment price growth) from Papanikolaou (2011), and aggregate per capita nondurables real consumption growth (ΔC). All these variables are naturally related to the overall state of the economy and are at least partially related to time-varying investment opportunities. As measures of the cost of debt (Group 2 in panel B) in the economy, we consider shocks to the tightness of collateral constraint on debt from Jermann and Quadrini (2012) (ΔJQ), a proxy for changes in the cost of external (equity plus debt) finance from Eisfeldt and Muir (2016) (ΔEM), the change in the BAA-AAA spread ($\Delta B-A$), and the risk-free rate (RF).

Panel B in Table 4 reports the previous correlations. The correlation between the equity ICS proxy and the alternative proxies of changes in investment opportunities is small (all correlations are between -8% and 17% and are not statistically significant at conventional levels). Similarly, the correlation between the equity ICS proxy and the various cost of debt (or debt and equity in ΔEM) measures investigated here is negative but small (all correlations are between -7% and -18% and are not statistically significant at conventional levels). These low correlations suggest that most of the variation in the equity ICS proxy in the data does not appear to be driven by time-varying investment opportunities nor by changes in the cost of debt over time, as captured by the variables considered here.

The Internet Appendix reports robustness tests that further expand the analysis in this section. To conserve space, we briefly summarize some of the results here, and refer the reader to the Internet Appendix for details. First, we consider an alternative equity ICS proxy computed as the component of the equity ICS proxy that is orthogonal to a broad set of variables correlated with time-varying investment opportunities and cost of debt (thus expanding the analysis of the univariate correlations described above). We show that the asset pricing tests obtained using this alternative orthogonal measure of equity ICS are similar to those reported below for the baseline measure. Second, we investigate the extent to which the equity ICS proxy captures information about systematic risk and financial conditions in the economy not captured by other macroeconomic variables used in the literature, some of which have already been shown to be correlated with asset prices in the cross-section. In particular, we use the component of equity ICS that is orthogonal to these variables.⁸ The conclusion is that the ability of the equity ICS proxy to price the cross-section of stock returns is not significantly affected once we control for the standard macroeconomic variables considered in the analysis. Finally, we show that the equity ICS proxy is positively correlated with other variables that are arguably positively correlated with equity ICS. For example, following Baker and Wurgler (2002 and 2006), we consider here shocks to market sentiment, because changes in market sentiment may affect the supply of equity capital, thus affecting the firms' effective cost of issuing equity. When market sentiment is high, investors might be more willing to supply equity capital to firms, making

⁸ Specifically, we orthogonalize (use regression residuals) the equity ICS to two sets of variables. The first set of variables includes measures closely related to financial shocks (broadly defined). We consider changes in the credit spread (BAA-AAA) and the risk-free rate; collateral constraint shocks from Jermann and Quadrini (2012); general external finance shocks from Eisfeldt and Muir (2016); time-varying aggregate liquidity shocks from Pastor and Stambaugh (2003); shocks to the balance sheet of financial intermediaries from Adrian, Moench, and Shin (2012) and Adrian, Etula, and Muir (2014); and changes in the capital ratio of financial intermediaries from He, Kelly, and Manela (2016). The second set of variables includes other primitive macroeconomic shocks that capture both systematic risk, aggregate risk premiums, and time-varying investment opportunities (which are all additional potential drivers of equity issuance decisions). We consider investment-specific technology shocks from Papanikolaou (2011); shocks to market sentiment from Baker and Wurgler (2002, 2006); uncertainty shocks from Bansal et al. (2014); the aggregate stock market return; and the Shiller's cyclically adjusted price-to-earnings ratio.

it effectively less costly for firms to issue equity (high equity ICS) (see McLean and Zhao 2014 for an empirical analysis of this link). We show that the equity ICS proxy is indeed positively correlated (correlation of 37% and statistically different from zero) with this alternative measure of equity ICS.

Taken together, the analyses in this section show that the equity ICS proxy is not contaminated in a significant way by time-varying investment opportunities nor by time-varying cost of debt (both of which can vary more in the real economy than in the model). This finding provides support to the interpretation that the equity ICS proxy captures time variation in the aggregate component of firms' cost of issuing equity in the real data, as implied by the model.

4. Main Findings

In this section, we investigate the link between equity ICS, systematic risk, and asset prices in the cross-section, both in the real data and in the model.

4.1 Methodology

We consider a two-factor asset pricing model with the aggregate stock market excess return and the equity ICS as the two factors. Specifically, we specify the following stochastic discount factor (investors' marginal utility):

$$M_t = 1 - b_M \times \text{MKT}_t - b_{ICS} \times \text{ICS}_t, \quad (21)$$

which states that investors' marginal utility is driven by two aggregate shocks, MKT, which is the market factor in the standard capital asset pricing model (CAPM), and the equity ICS proxy. This specification of the stochastic discount factor is closely related to that in the model given in Equation (15) because the market factor is used here as a proxy for the aggregate productivity shock.⁹

We then estimate the risk factor loadings on the two aggregate shocks (b_M and b_{ICS}) by the generalized method of moments (GMM) using the standard asset pricing moment condition $E[r_{it}^e M_t] = 0$, in which r_{it}^e is the excess return of test asset i , in year t . To help in the interpretation of the results, we can write this moment condition as:

$$E[r_{it}^e] = \alpha_i + b_M \text{Cov}(\text{MKT}_t, r_{it}^e) + b_{ICS} \text{Cov}(\text{ICS}_t, r_{it}^e), \quad (22)$$

where we added the term α_i (alpha), the pricing error (interpreted as an abnormal return) associated with asset i . For comparison, we also estimate the one-factor CAPM (which corresponds to the restricted case in which $b_{ICS} = 0$ in Equation 22), to establish the marginal importance of the equity ICS for capturing

⁹ In the model, most of the variation in the aggregate stock market return is driven by the aggregate productivity shock. Specifically, across panels, a multivariate time-series regression of the aggregate stock market return on the two risk factors has an average regression $R^2 \approx 98\%$, a univariate regression on the aggregate productivity shock has an average regression $R^2 \approx 87\%$, but a univariate regression on the aggregate equity issuance cost shock has an average regression $R^2 \approx 11\%$ (results not tabulated).

systematic risk in the economy. In addition, it is well known that the CAPM does not work well in practice, so we use the failure of the CAPM as an additional empirical fact in the evaluation of the theoretical model.

As test assets, we focus the analysis in the model on 5 book-to-market and 5 investment rate portfolios. These portfolios are closely linked to investment, and hence are well aligned with the neoclassical model of investment that we investigate here. To enhance the precision in the measurement of the risk factor loadings (especially given the relatively small number of test assets in the benchmark portfolios) and establish the importance of the equity ICS for asset prices in the economy, we also consider a larger set of test assets in the real data. Following Lewellen, Nagel, and Shanken (2010), in addition to the book-to-market and investment portfolios, we consider 17 Fama-French industry portfolios to break the strong factor structure of the baseline portfolios. We also consider 25 portfolios double sorted on book-to-market and size, as well as 25 portfolios double sorted on investment and size. That is, in the larger set of test assets, we consider a total of 67 portfolios. Appendix A-3 explains the construction of these portfolios. To be conservative and save space, all portfolio returns are value-weighted, but we note that the results reported here are, in general, stronger when we use equal-weighted portfolio returns.

4.2 Equity ICS and systematic risk

In the model, the equity ICS is a source of systematic risk because it affects the stochastic discount factor (investor's marginal utility). In addition, the stochastic discount risk factor loading parameter on the equity ICS (γ_ξ) is positive, which implies a positive price of risk for this shock. Here, we test these key model assumptions in the real data using the equity ICS proxy.

Table 5 reports the GMM estimates of the risk factor loadings, and implied mean absolute pricing errors (MAE), obtained from the estimation of the asset pricing model given in Equation (22) using different sets of test assets in the real data. The table shows that the estimated risk factor loading on the equity ICS proxy (and hence the equity ICS price of risk) is positive, and is significantly different from zero across all the different sets of test assets considered here (all estimates of the equity ICS risk factor loading are more than 2 standard errors from zero). The estimates of the equity ICS risk factor loading range from $b_{ICS}=8$, when all 67 test assets are considered, to $b_{ICS}=16$ when the five book-to-market portfolios are considered.

In addition, Table 5 shows that the two-factor model performs significantly better than the CAPM in explaining the returns of these portfolios. When the equity ICS factor is added to the market factor, the MAE decreases significantly relative to the MAE of the CAPM. For example, across the five book-to-market portfolios, the MAE drops from 2.1% per annum in the CAPM to 0.7% per annum in the two-factor model.

Taken together, the results in this section show that the equity ICS proxy is correlated with systematic risk in the economy, and has a positive price of

Table 5
Price of risk of equity issuance cost shocks

	5 BM		5 IK		50 SZ-BM-IK		67 SZ-BM-IK and industry	
	CAPM	2F	CAPM	2F	CAPM	2F	CAPM	2F
b_M	2.68	1.24	2.34	1.21	2.76	1.41	2.68	1.77
$[t]$	3.52	0.67	2.93	0.70	2.19	0.85	1.86	1.00
b_{ICS}		15.74		14.89		11.65		8.04
$[t]$		2.23		1.86		2.46		2.17
MAE	2.08	0.65	1.79	1.02	2.16	1.24	2.12	1.60

This table reports the estimates of the risk factor loadings and pricing errors of two asset pricing models: the CAPM, in which the return on the market (MKT) is the only pricing factor, and a two factor model (2F), in which the return on the market and the equity ICS proxy (ICS) are the two factors. Each model is estimated by the generalized method of moments (GMM) using the standard asset pricing moment condition $E[r_{it}^e M_t] = 0$, in which $M_t = 1 - b_M \times \text{MKT}_t - b_{ICS} \times \text{ICS}_t$ is the model specific stochastic discount factor (SDF), MKT_t is the (demeaned) market return, ICS_t is the (demeaned) equity ICS proxy, and b_M and b_{ICS} are the corresponding risk factor loadings on the SDF. $[t]$ is the corresponding factor t-statistic. The estimation of the asset pricing model is performed on four sets of test assets: five book-to-market portfolios (5 BM), five investment rate portfolios (5 IK), 25 5-Size-5-BM portfolios plus 25 5-Size-5-IK portfolios (50 SZ-BM-IK), and the previous 50 portfolios plus 17 Fama-French industry portfolios (67 SZ-BM-IK and Industry). MAE is the estimation implied mean absolute pricing errors (mean of pricing errors, a , inferred from the errors on the moment condition estimated above). Data are annual from 1964 to 2013.

risk. Thus, consistent with the baseline calibration of the model, periods when issuing equity is particularly costly (low equity ICS) tend to occur at times when investor's marginal utility is high, that is, are bad economic times.

4.3 Equity ICS and asset prices

Table 6 reports the average excess returns, Sharpe ratios (average excess return-to-excess return standard deviation ratio), the multivariate covariances of the portfolio returns with the two risk factors, and the CAPM and the two-factor model-implied pricing errors (α and α^{2F} , respectively) of the low (L), high (H), and the spread (H-L) portfolios across the book-to-market and investment portfolio sorts in both the real data (columns under Data) and in the model (columns under Model). To obtain the results in the model, we replicate the empirical procedures using artificial data obtained from the simulation of the model.

Table 6 (Data) shows that, consistent with previous studies, value firms (high book-to-market, H) in the real data outperform growth firms (low book-to-market, L) by about 6.5% per annum, and also have a higher Sharpe ratio, 0.60 versus 0.28, respectively. The CAPM pricing error of the high-minus-low (H-L) book-to-market portfolio is higher than the average return spread itself, 7.2% per annum, and this value is more than 2.0 standard errors from zero. The pricing error of this spread portfolio drops to close to zero in the two-factor model, when the equity ICS is included ($\alpha^{2F} = 0.2\%$).

Similarly, Table 6 shows that, consistent with previous studies, firms with currently high investment rates in the real data underperform firms with currently low investment rates by about -4% per annum, and also have a lower Sharpe ratio, 0.22 versus 0.50, respectively. The CAPM pricing error of

Table 6
Equity issuance cost shocks and asset prices in the cross-section

	Data						Model					
	5 BM			5 IK			5 BM			5 IK		
	L	H	H-L	L	H	H-L	L	H	H-L	L	H	H-L
Portfolio returns and Sharpe ratios												
$E[r^e]$	5.76	12.30	6.54	9.40	5.38	-4.02	3.66	9.37	5.71	9.50	3.68	-5.81
$[t]$	2.45	6.13	2.98	4.19	2.27	-2.05	4.98	13.69	25.36	13.45	5.17	-37.54
SR	0.28	0.60	0.40	0.50	0.22	-0.25	0.18	0.48	1.06	0.48	0.18	-1.43
Risk factor betas												
$[r]$												
Cov^{MKT}	3.44	3.09	-0.35	2.93	3.92	0.99	3.70	3.50	-0.2	3.56	3.62	0.06
$[t]$	5.74	5.03	-0.84	5.59	4.84	1.81	15.52	15.24	-4.68	15.32	15.57	1.89
Cov^{ICS}	-0.18	0.27	0.45	0.25	-0.13	-0.38	-0.11	0.11	0.22	0.10	-0.10	-0.2
$[t]$	-2.27	3.06	3.61	3.16	-1.21	-2.27	-3.37	2.91	10.45	2.64	-3.25	-10.52
Pricing errors: CAPM and 2-factor model												
α	-3.36	3.87	7.23	2.41	-3.74	-6.15	-1.89	4.12	6.01	4.15	-1.76	-5.9
$[t]$	-1.91	2.06	2.02	1.65	-2.30	-2.08	-8.62	17.33	27.22	18.59	-8.69	-42.48
α^{2F}	0.71	0.86	0.15	-0.83	-1.32	-0.49	0.08	-0.06	-0.14	0.03	0.21	0.18
$[t]$	0.95	0.78	0.23	-1.28	-1.30	-1.12	0.70	-0.59	-1.65	0.23	1.62	1.45

This table reports in both the real data (Data columns) and in the model (Model columns) the average value-weighted return characteristics of two sets of portfolio sorts: five book-to-market portfolios (5 BM), and five investment portfolios (5 IK). For each portfolio sort, the table reports the characteristics of the portfolios 1 (Low, L), and five (High, H), and the high-minus-low portfolio (H-L). $E[r^e]$ is the portfolio average annual excess return; $[t]$ is the corresponding Newey-West t -statistic; SR is the portfolio Sharpe ratio; Cov^{MKT} is the multivariate covariance between the portfolio excess return and the MKT factor; and Cov^{ICS} is the multivariate covariance between the portfolio excess return and the ICS factor. α and α^{2F} are the pricing errors (abnormal returns) implied by the estimation of the CAPM and the two-factor model, respectively. The pricing errors are inferred from the errors on the moment condition estimated in Table 5. Real data are annual from 1964 to 2013. The reported statistics for the model are obtained from 100 samples of simulated data, each with 3,600 firms and 600 monthly observations. Returns are then aggregated to annual frequency to match the frequency of the real data.

the high-minus-low investment rate portfolio is large, -6.2% per annum, and this value is more than 2.0 standard errors from zero. Again, the pricing error of this spread portfolio drops to close to zero in the two-factor model, when the equity ICS is included ($\alpha^{2F} = -0.5\%$).

To help understand the source of the improved performance of the two-factor model relative to the CAPM in pricing these portfolios in the real data, Table 6 also reports the multivariate covariances of the portfolio returns with the two risk factors. The covariance of the portfolio returns with the equity ICS is increasing across the book-to-market portfolios, and decreasing across the investment portfolios. That is, the value/low investment firms are more sensitive (higher covariance) to changes in the cost of external equity financing, as reflected in their relatively higher covariance with the equity ICS. Because low realizations of the equity ICS are high marginal utility states, this makes the value/low investment firms riskier, which explain the higher average returns of these firms in equilibrium.

Turning to the analysis of the model results, Table 6 (Model) shows that calibration of the baseline model generates a pattern of average excess returns across the book-to-market portfolios that is similar to the pattern in the data. Value firms outperform growth firms by about 5.7% per annum, and also have a

higher Sharpe ratio, 0.48 versus 0.18, respectively. These values are comparable with those in the data. We note, however, that the Sharpe ratio of the book-to-market spread portfolio is larger than in the data.

Like in the real data, in the model, the CAPM is unable to capture the value spread. The pricing error of the high-minus-low (H-L) book-to-market portfolio is 6% per annum, which is close to the 7.2% per annum observed in the real data. The pricing error of this spread portfolio drops to close to zero in the two-factor model, when the equity ICS is included ($\alpha^{2F} = -0.1\%$). Note that to make the comparison with the empirical results meaningful, we perform the asset pricing tests in the model using the model-implied equity ICS proxy (by estimating Equation (20) in the model, and using the fitted values of the regression). Naturally, given the high correlation (93%) between the true equity ICS and its proxy in the model, the asset pricing test results are almost identical to those obtained using the true equity ICS.

The model also matches the pattern of average returns of the investment portfolios. Consistent with the data, firms with currently high investment rates underperform firms with currently low investment rates by about -5.8% per annum, and also have a lower Sharpe ratio, 0.18 versus 0.48, respectively. The CAPM pricing error of the high-minus-low investment rate portfolio is large, -5.9% per annum, which is close to the -6.15% per annum observed in the real data. The pricing error of this spread portfolio drops to close to zero in the two-factor model, when the equity ICS is included ($\alpha^{2F} = 0.18\%$).

The value and investment return spreads in the model are mostly driven by firms' differential exposure to the equity ICS. To show this result in a clear manner, we report the covariances of the portfolio returns with respect to the two aggregate risk factors in the model in Figure 2, for all book-to-market and investment portfolios. This plot complements the information in Table 6 which reports the covariances of the low and high portfolios with the two risk factors, together with the formal test for the difference in these covariances. To highlight the cross-sectional dispersion in the portfolios' exposure to the shocks, the figure shows a scaled return covariance; that is, the portfolio's return covariance with each factor relative to the corresponding covariance of the first portfolio.

The two left panels in Figure 2 show that the sensitivity of the returns of the book-to-market and the investment portfolios to the market factor (aggregate productivity shock) is almost flat across the portfolios, which is the source of the failure of the standard CAPM in capturing the difference in the returns of these portfolios in the model. In contrast, the dispersion in the sensitivity to the aggregate equity ICS is large, and it is monotonically increasing across the book-to-market portfolios, and monotonically decreasing across the investment portfolios. This differential exposure is the fundamental difference in the quantity of risk of the book-to-market and investment portfolios in the model, and explains why the value and low investment firms have higher average returns in equilibrium.

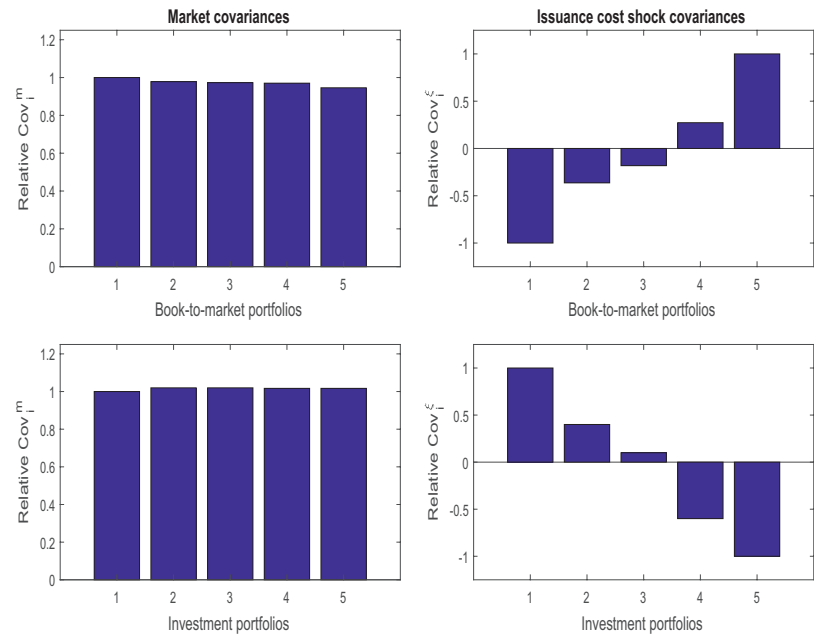


Figure 2
Market and equity issuance cost shock covariances
This figure reports the risk exposures (covariances) of the excess returns of the five book-to-market and five investment portfolios, with respect to the aggregate market factor, and the aggregate equity issuance cost shock factor, using data simulated from the model. The covariances are expressed relative to the covariance of the first (low) portfolio in each sort to emphasize the cross-sectional variation. The reported statistics for the model are obtained as averages from 100 samples of simulated data, each with 3,600 firms and 600 monthly observations.

Taken together, the model generates sizeable value and investment return spreads due to differential portfolios’ exposure to the equity ICS factor, not by differential portfolios’ exposure to the CAPM market factor (aggregate productivity shock), consistent with the data.

4.4 Investment, financial flows, and other characteristics across portfolios

The differential exposure of the book-to-market and investment portfolios to the equity ICS reflects the differences in the characteristics of the firms in these portfolios. To explain these differences, and evaluate the extent to which the model is consistent with them, Table 7 reports for each portfolio sort selected characteristics of the firms in the low (L), high (H), and the spread (H-L) portfolio in both the real data (Data columns) and in the model (Model columns).

We focus on the following firm characteristics that characterize the value, investment, financial flows (equity and debt financing), capital structure, and productivity of the firms in each portfolio at the time of portfolio formation: book-to-market ratio (BM), market value of equity (size), investment rate in

Table 7
Portfolio characteristics in the data versus model

	Data						Model					
	5 BM			5 IK			5 BM			5 IK		
	L	H	H-L	L	H	H-L	L	H	H-L	L	H	H-L
Book-to-market	0.26	1.88	1.62	0.86	0.68	-0.17	0.14	0.30	0.16	0.28	0.15	-0.13
Investment	0.35	0.16	-0.19	0.11	0.39	0.28	0.30	0.00	-0.30	-0.06	0.50	0.56
Size	4.69	3.52	-1.17	4.06	4.72	0.66	5.13	5.12	-0.01	4.96	5.17	0.21
Equity/assets	0.42	0.07	-0.34	0.30	0.31	0.01	0.15	0.11	-0.04	0.05	0.21	0.16
Δ Debt/assets	0.03	0.00	-0.03	-0.01	0.05	0.06	0.25	-0.21	-0.46	-0.33	0.92	1.25
Leverage	0.26	0.26	0.00	0.29	0.22	-0.06	0.16	0.30	0.14	0.27	0.17	-0.10
TFP	0.95	0.65	-0.30	0.70	0.87	0.16	1.79	1.13	-0.66	0.86	1.52	0.66

This table reports average portfolio characteristics of five book-to-market (5 BM) and five investment rate (5 IK) portfolios in the real data (Data columns), and in the data simulated from the model (Model columns). For each portfolio sort we report the characteristics of portfolios 1 (Low, L), and five (High, H). H-L stands for the high-minus-low portfolio. Book-to-market ratio is the book value-to-equity market value of the firm; Investment is the physical capital investment rate; Size is log of ex dividend firm value; Equity/assets is the gross equity issuance-to-asset ratio; Δ Debt/assets is the change in total debt-to-assets ratio; Leverage is the firms' book leverage ratio; and TFP is firms' total factor productivity (TFP), a measure of productivity (in the model, $TFP = \log(Z)$, and in the real data the firm-level TFP is from Tuzel and Imrohorglu, 2014). The portfolio-level characteristic is the time-series average of the average characteristic across the firms in the portfolio in each year. We winsorize the variables at top and bottom 2% when computing the portfolio average characteristics in the data. Data are annual from 1964 to 2013. The reported statistics for the model are obtained from 100 samples of simulated data, each with 3,600 firms and 600 monthly observations.

physical capital (IK), gross equity issuance-to-asset ratio (Equity/Assets), debt growth (Δ Debt/Assets), leverage ratio, and firm-level total factor productivity (TFP). Appendix A-3 describes how these characteristics are computed in the data. We construct the average characteristics for each portfolio by first computing the mean of each characteristic across all firms in the portfolio in a given year, and then report the corresponding time-series average.

Table 7, column Data, shows that the qualitative relationship between the characteristics of these firms and its level of risk (average returns) is consistent across the book-to-market and the investment portfolios. In general, low risk (growth, L, and high investment, H) firms have a lower book-to-market ratio, are bigger, invest more, issue more debt and equity, are less levered, and are more productive than high risk (value, H, low investment, L) firms. Table 7, column Model, shows that the model qualitatively matches the pattern of the characteristics of these portfolios.

5. Inspecting the Mechanism

In this section we perform several analyses to understand the economic mechanism driving the overall good fit of the model.

5.1 The driver of risk premiums in the cross-section

According to the analysis reported in Section 4.3, the average return spreads in the model are driven by the differential exposure of the returns of portfolios to the aggregate equity ICS, not by differential exposure to the market factor (aggregate productivity shock). Why do the returns of firms with currently high

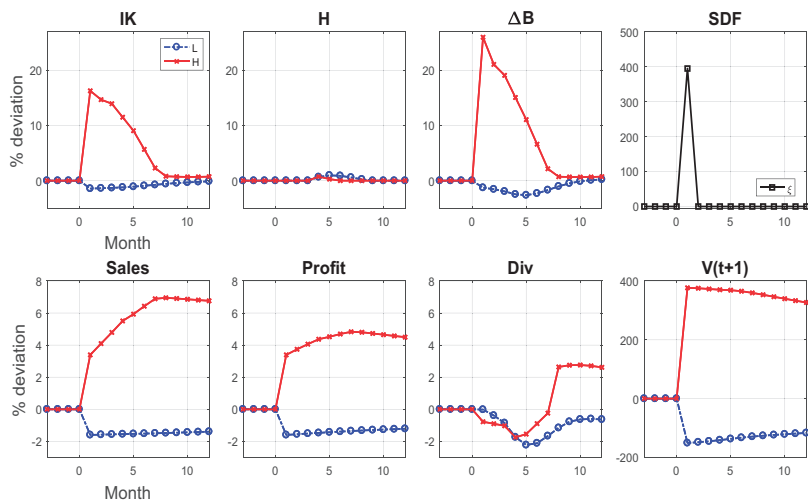


Figure 3
Impulse response to a negative equity issuance cost shock

Impulse responses of selected endogenous variables in the baseline calibration of the model to a one standard deviation negative aggregate equity ICS (higher cost of issuing equity, bad times). The responses are measured as a percentage of the deviation relative to the long-run average values (time detrended, when applicable). To generate the response of a high productivity (H) firm, we add a positive 1 standard deviation firm-specific productivity shock. To generate the response of a low productivity firm (L), we add a negative 1 standard deviation firm-specific productivity shock. The frequency of the data is monthly. IK is firms' investment rate; ΔB is firms' debt change; SDF is the stochastic discount factor (consumers' marginal utility); Sales is measured as output Y ; Profits is after tax corporate profits; Div is firms' dividends; and V is the continuation value of the firm (price of the firm after dividends).

book-to-market ratio and low investment rates, have higher positive covariance with the aggregate equity ICS in equilibrium? Given the positive price of risk of this shock, understanding this endogenous covariance is essential for understanding the equilibrium return spreads in the model.

To explain the economic mechanism behind the results reported in the previous sections, Figure 3 shows impulse responses of selected endogenous variables in the baseline calibration of the model to a one standard deviation negative aggregate equity ICS, ξ (i.e., an increase in the marginal cost of equity financing, bad times). We report the responses of each variable relative to its (time-detrended) long-run average level. Because all firms in the economy are *ex ante* identical, we generate cross-sectional heterogeneity by examining the response of two firms in which their respective firm-specific productivity level is set one standard deviation above and below the long-run average level of firm productivity (we label these two firms as high and low productivity firms, respectively); furthermore, their productivity levels gradually mean revert to the average level following Equation (3).¹⁰ The high and low productivity firms

¹⁰ The long-run average level is determined by setting all shocks to the long-run average level, that is, $z = -3.9$, $\xi = 0$, and $\Delta x = 0$.

correspond roughly to the growth/high investment and value/low investment firms in the model, respectively. Even though the difference in productivity is not the only difference across these firms, it is clearly an important state variable that varies systematically across these portfolio as reported in Table 7. This is because idiosyncratic productivity is the underlying primitive source of cross-sectional heterogeneity in the model.

Figure 3 shows that after a negative equity ICS, the high productivity firms still want to increase their investment while the low productivity firms want to decrease it. Because of the increase in the marginal cost of external equity financing, the external equity market freezes upon impact for both firms for an extended period of more than three months. The increase in investment and corresponding adjustment costs of the high productivity firms is financed by an increase in debt. The high productivity firms can rely on debt financing because they are accumulating more capital which allows them to pledge for more debt. For low productivity firms, debt falls substantially for an extended period of time because these firms lack both internal funds (sales fall) and external funds (external equity is costly) to pay off the debt and hence they have to deleverage. The dividends of the high productivity firms fall slightly upon impact and then increase immediately above the long-run average level for a long period of time; the dividends of low productivity firms fall immediately below the steady state level for an extended period of time. As a result of the response of firms' profits and dividends over time, the continuation value (the present value of all future dividends at time $t + 1$) of the high productivity firm still increases on impact despite the negative equity ICS, but the continuation value of the low productivity firm decreases (relative to its long-run average level) on impact. Because current dividends represent a small fraction of total firm value, the properties of firm-level stock returns are mostly determined by the change in the continuation value, the standard capital gains component of stock returns. Given that the high productivity firms are less affected by the equity ICS, the returns of these firms have a relatively lower covariance with the equity ICS, while the returns of the low productivity firms have a relatively higher covariance with the equity ICS. In turn, because the stochastic discount factor (marginal utility) is decreasing in this shock, the differential covariance implies that, all else equal, the high productivity (growth, high investment) firms have relatively lower risk than low productivity (value, low investment) firms.

5.2 Role of positive price of risk in equity ICS

According to Equation (19), the impact of the firm's differential exposures to the aggregate equity ICS on equilibrium risk premiums depends on the price of risk of this shock, which is determined by the (exogenously specified) stochastic discount factor risk factor loading on this shock (γ_ξ). To evaluate the importance of this parameter for the model's results, we perform comparative statics with respect to this parameter.

Table 8
Selected moments in the data and across alternative calibrations of the model

Spec #	Quantities				Asset prices				
	S.D.	Correl.	Correl.	Cash/asset	MKT	BM		IK	
	IK	(ICS, $\Delta \xi$)	(ICS, ΔSales)		r^e	r^e	α	r^e	α
0-Data	0.24	na	0.11	0.05	5.71	6.54	7.23	-4.02	-6.15
1-Baseline	0.22	0.93	0.11	0.05	5.58	5.71	6.01	-5.81	-5.90
2-Zero price of risk of equity ICS ($\gamma_\xi = 0$; baseline $\gamma_\xi = 12$)	0.28	0.76	0.13	0	12.2	-1.66	1.09	-0.59	-1.80
3-Zero issuance cost ($\eta_0 = \eta_1 = \eta_2 = 0$)	0.34	0	0.50	0.02	5.64	-0.51	0.04	-1.29	-0.33
4-No stochastic equity issuance cost ($\eta_3 = 0$)	0.32	0.10	0.25	0.02	6.43	-0.33	-0.13	-1.22	-0.70
5-Tight collateral constraint ($\varphi = 0.3$; baseline $\varphi = 0.85$)	0.21	0.76	0.12	0.07	6.23	-2.52	-1.61	1.22	0.71
6-High debt adjustment cost ($c_b = 6$; baseline $c_b = 0.3$)	0.24	0.72	0.34	0.10	5.14	1.29	2.65	-2.03	-2.13

This table presents several comparative statics exercises across six specifications (spec #). The reported statistics for the model are obtained from 100 samples of simulated data, each with 3,600 firms and 600 monthly observations.

Table 8 reports selected model-implied moments from several alternative specifications of the model, which we compare against the moments in the data (specification 0) and in the baseline calibration of the model (specification 1). In specification 2, we set the stochastic discount factor risk factor loading on the equity ICS to be zero ($\gamma_\xi = 0$ vs. $\gamma_\xi = 12$ in the baseline model). In this specification, we keep all the other model parameters equal to the baseline specification.

Specification 2 in Table 8 shows that decreasing the size of the stochastic discount factor risk factor loading on the aggregate equity ICS has a relatively small effect on the properties of firms' investment rates, slightly increasing its volatility from 22% in the baseline model to 28% here. But the impact on asset prices is large. All the return spreads drop substantially. The value spread decreases from 5.7% in the baseline model to -1.7% here, and the investment return spread decreases (in absolute value) from -5.8% in the baseline model to -0.6% here. This analysis shows that, consistent with the empirical results reported in Section 4.2, a sufficiently large and positive risk factor loading on the equity ICS is important for the model to generate enough risk dispersion in the cross-section, that is, sizeable portfolio return spreads.¹¹

¹¹ Decreasing the size of the stochastic discount factor risk factor loading on aggregate productivity shock also has a relatively small effect on quantities, slightly reducing the volatility of the firms' investment rate. The risk premium on the aggregate stock market is significantly reduced. However, the value and the investment return spreads remain sizable and comparable in magnitude with the corresponding spreads in the baseline calibration. This result thus confirms that the aggregate productivity shock drives the aggregate market premium but not the variation in risk premiums in the cross-section.

5.3 Role of positive and time-varying equity issuance costs

The existence of positive and time-varying external equity issuance costs is important for the overall good fit of the model. To show this importance, we compute the model-implied moments from alternative calibrations of the equity issuance cost function, which we report in Table 8. In specification 3, we shut down equity issuance costs completely ($\eta_0 = \eta_1 = \eta_2 = 0$). In specification 4, we shut down variation in equity issuance costs over time ($\eta_3 = 0$) (we keep the effect of the equity ICS on the stochastic discount factor, thus allowing this shock to still affect asset prices through this channel).¹²

Specification 3 in Table 8 shows that, by removing equity issuance costs, the model generates a firm-level investment rate that is too volatile (22% in the baseline model and the data vs. 34% here). Furthermore, the firm-level correlation of equity issuance with sales growth is too large relative to the data (0.11 in the baseline model and the data vs. 0.5 here), because here the equity issuance decision is primarily driven by productivity shocks. This result is consistent with Eisfeldt and Muir (2016), who also show that external financing shocks helps reduce the correlation of issuance and sales growth. The effect of removing equity issuance costs on asset prices is also large. The value and the investment return spreads reduce significantly relative to the baseline model from 5.7%/−5.8% in the baseline model, respectively, to −0.5%/−1.3%, respectively, here.

Turning to the analysis of the impact of time-varying equity issuance costs on the model results, specification 4 shows that removing the time variation in equity issuance costs generates qualitative effects that are very similar to those reported in the previous specification 3. In particular, this specification of the model generates a firm-level investment rate that is too volatile (22% in the baseline model and the data vs. 32% here). In addition, the firm-level correlation of equity issuance with sales growth is too large relative to the data (0.11 in the baseline model and the data vs. 0.25 here). The value and the investment return spreads are all significantly reduced (in absolute value) from 5.7%/−5.8%, respectively, in the baseline model, to −0.3%/−1.2%, respectively, here.¹³

5.4 Role of substitution between equity and debt financing

According to the analysis of the impulse response functions in Section 5.1, the economic mechanism to generate risk dispersion in the model hinges

¹² We change γ_x to 1.5 and 2 in specifications 3 and 4, respectively, so that the market returns are comparable to that of the baseline calibration.

¹³ Note that capital adjustment costs also matter quantitatively for risk premiums in the cross-section (in general, lowering capital adjustment costs decreases the cross-sectional risk dispersion). The real frictions (costly reversibility) are necessary to match the firms' investment rate dynamics. But, as we show here, the positive and time-varying equity issuance costs (and associated equity ICS) are crucial for the model to simultaneously generate the large variation in risk premiums in the cross-section as well as the failure of the CAPM in explaining this variation.

on the ability of the more productive firms to substitute debt financing for equity financing when the cost of issuing equity increases. Specifically, it is the flexibility of the productive firms in switching between marginal sources of financing during bad economic times that makes these firms to be less risky.

To further understand the quantitative importance of the previous mechanism, Table 8 reports selected model-implied moments from alternative calibrations of the collateral constraint and debt adjustment costs. In specification 5, we tighten the collateral constraint by setting the resale value of capital to $\varphi=0.3$ instead of $\varphi=0.85$ in the baseline model. In specification 6, we increase the size of the debt adjustment costs by twenty times relative to the baseline calibration, $c_b=6$ instead of $c_b=0.3$ in the baseline model.

Table 8 shows that all the return spreads decrease substantially in specifications 5 and 6. This happens because tightening the collateral constraint or increasing the debt adjustment cost raises the effective cost of using and changing debt, causing more firms to save in cash due to a precautionary saving motive. The channel available for more productive firms to substitute debt for equity financing in low equity ICS states (baseline specification) is effectively turned off. Thus, in these specifications, both high and low productive firms use cash to smooth the impact of a negative equity ICS, significantly reducing the endogenous cross-sectional risk dispersion in the model.

Given the importance of the debt-equity substitution mechanism for the good fit of the model, we test this mechanism in the real data. Table 9 reports the financial flows of the book-to-market and investment portfolios across the financial cycle. To define the financial cycle, we split the sample into low, medium, and high equity ICS states based on the bottom and top 10 percentiles of the time-series distribution of the equity ICS proxy. We then compute, within each ICS state, the time-series average of the portfolio-level realized (i.e., after portfolio formation and hence contemporaneous with the realized equity ICS) fraction of equity issuance (Equity issuance fraction), cross-sectional mean gross equity issuance-to-asset ratio (Gross equity issuance/Assets) computed among the set of firms with positive issuance, and the cross-sectional mean of the changes in debt over assets ($\Delta\text{Debt}/\text{Assets}$), for the high (H) and low (L) portfolios in each sort. To remove the effects of changes in market conditions on the equity and debt decisions unrelated to changes in the financial cycle (equity ICS), we orthogonalize these variables to changes in aggregate gross domestic product (GDP)¹⁴

¹⁴ We obtain these components by first running a regression of the portfolio-level moments on a constant and aggregate real GDP growth. We then compute, for each portfolio and for each ICS state, the time-series average of the intercept plus the regression residual of the previous regression.

Table 9
Financial flows across the financial cycle

	5 BM		5 IK	
	L	H	L	H
		Equity issuance fraction		
High ICS, good times	30.19	11.38	23.16	22.56
Mid	23.97	8.26	16.92	17.05
Low ICS, bad times	22.41	7.76	14.36	13.54
		Gross equity issuance/assets		
High ICS, good times	55.11	9.66	42.45	40.13
Mid	42.62	7.10	32.34	31.53
Low ICS, bad times	39.82	6.82	30.67	31.08
		Δ Debt/assets		
High ICS, good times	0.29	-1.18	-1.82	1.28
Mid	2.39	-0.10	-0.82	3.02
Low ICS, bad times	2.74	-0.39	-0.72	3.34

This table presents the time-series average of portfolio-level realized (i.e., in the year after portfolio formation) fraction of firms in the portfolio that issue equity (Equity issuance fraction), gross equity issuance-to-asset ratio (Gross equity issuance / assets) computed among issuing firms, and the change in debt over assets (Δ Debt/assets) for the low (L) and high (H) portfolios sorted on book-to-market (BM) and investment (IK). Among them, the portfolio-level Gross equity issuance/assets and Δ Debt/assets are computed as the averages across all firms in a portfolio. We winsorize the firm variables at top and bottom 2% to remove the outliers when computing the averages. All numbers reported are in percentage. The reported time-series averages of the portfolio characteristics are computed separately across different equity ICS states, defined as periods characterized by high, medium, and low realizations of the model-implied equity ICS in the real data. The High ICS state periods (good times), are defined as the years in which the realized equity ICS is in the top tenth percentile of the ICS distribution. The low ICS states (bad times), are defined as the years in which the realized ICS is in the bottom tenth percentile of the ICS distribution. The intermediate (Mid) states corresponds to the years in which the realized ICS is between the 10th and 90th percentiles of the equity ICS distribution. The reported variables are the components of the variables that are orthogonal to GDP growth to remove the effect of changes in market conditions related to GDP. Data are annual from 1964 to 2013.

Consistent with the debt-equity substitution mechanism in the model, Table 9 shows that low risk (growth/high investment) firms issue less equity and increase their debt when equity becomes more costly (low equity ICS states). Specifically, as we move from the high equity ICS state (low cost of issuing equity) to the low equity ICS state (high cost of issuing equity), we see that the amount of equity issuance and the fraction of firms issuing equity goes down across all firms. At the same time, for low risk (growth/high investment) firms, the debt change goes from close to zero, to significantly positive. That is, low risk (growth/high investment) firms substitute equity financing for debt financing when external equity financing is too costly, consistent with the model predictions.

For high risk (value/low investment) firms, we note that these firms in general want to reduce their debt levels (debt change is negative) across all equity ICS states as they are downsizing. The increase in the equity issuance cost reduces the speed with which these firms can shed off their debt. This effect in turn constrains the financial flexibility of these firms by making it more difficult for them to issue equity to pay back their debt (which is what these firms would like to do if issuing equity was free or not particularly costly). That is, high risk (value/low investment) firms are unable to substitute equity financing for

debt financing when external equity financing is too costly, consistent with the model predictions.¹⁵

5.5 Internal validation of the equity ICS proxy

We can use the alternative calibrations of the model to validate the model-implied proxy of equity ICS. Because in the model we observe both the true equity ICS ($\Delta\xi$) and its model-implied proxy (from Equation 20), we can investigate the conditions under which the true equity ICS and its proxy are significantly positively correlated. As we show here, the link is not mechanical, and depends on key model features.

To examine this question, column Correl (ICS, $\Delta\xi$) in Table 8 reports the correlation between the true equity ICS and its model-implied proxy across the different calibrations of the model. Several interesting conclusions emerge from this exercise. First, as noted, the correlation between the true equity ICS and its proxy in the benchmark model is very high, 93%, which is the evidence we use in Section 3.1 to justify the use of the model-implied proxy in the empirical analysis.

In addition, the sizeable and positive correlation between the true equity ICS and its model-implied proxy relies on the existence of both time-varying (stochastic) and positive equity issuance costs. In specifications 3 and 4, when we either shut down the equity issuance cost or its time variation (keeping the effect of the equity ICS on marginal utility, and hence, still allowing this shock to affect asset prices), the correlations between the true equity ICS and its proxy are close to zero. Thus, under these specifications, the true equity ICS is not identified from the regression in Equation (20) (i.e., the regression R^2 is essentially zero; not tabulated). This happens because in these specifications, the variation in the fraction of firms issuing equity in the cross-section and leverage ratios in the model are entirely driven by the aggregate productivity shock, not by the equity ICS.

Furthermore, in specification 2, when the price of risk of the equity ICS is set to zero, the correlation between the true equity ICS and its proxy drops to 0.76 compared to the benchmark, but is still sizable. This high correlation implies that it is the cash flow channel (impact of equity ICS on firms' cash flows) that contributes the most to the identification of the equity ICS through Equation (20) in the model.

¹⁵ We also find that low risk (growth/high investment) firms do not reduce their cash holding to finance their operations with internal funds in periods in which it is costly to issue equity (see the Internet Appendix). Indeed, in low equity ICS states (high cost of issuing equity), these firms are all still accumulating cash, although at a much smaller rate than during high equity ICS states. High risk (value/low investment) firms reduce their cash holdings in high equity ICS states, but they also de-leverage, as reported here. These results suggest that the effect of reducing cash for high risk firms is not nearly enough to smooth their payout even if they also reduce their existing debt. This result suggest that the more empirically relevant margin to explain the value and the investment return spreads is the choice of debt versus equity when equity ICS is low, whereas the choice between cash and equity does not appear to be as important.

Taken together, the ability of Equation (20) (and hence of the model-implied equity ICS proxy) to properly identify the true equity ICS in the model relies on positive and time-varying external equity issuance costs that affect firms' cash flows.

5.6 Time-varying cost of issuance equity or debt?

Finally, we also consider an extended version of the model with time-varying costs of issuing both equity and debt, to investigate the relative importance of time-varying frictions in external equity versus debt financing for our findings.

In the model, equity is the only source of external financing with time-varying issuance costs. Previous research (e.g., Jermann and Quadrini 2012 and others, as discussed in Section 1), documents the importance of time-varying frictions in the debt market for explaining the dynamics of several macroeconomic variables and asset prices. To understand how our model generates large value and investment return spreads even without time-varying frictions in the debt market, we perform two analyses. First, theoretically, we consider an extended version of the model where we add an aggregate debt issuance cost shock (henceforth debt ICS) and investigate its implications for asset prices. Second, empirically, we examine the relationship between our equity ICS proxy, a proxy of debt ICS, and the link between the debt ICS proxy and asset prices in the cross-section.

5.6.1 Role of debt ICS in the extended model. Following Jermann and Quadrini (2012), we introduce a debt ICS in the model by specifying the tightness of the collateral constraint parameter to be time-varying (φ_t). The variation in this parameter is driven by an aggregate debt ICS. This shock also affects the stochastic discount factor and has a positive price of risk, in an analogous fashion to the specification of the equity ICS in the baseline model. We consider a version of the model in which the debt ICS is the only financial shock (and hence shut down equity ICS), and also a version of the model with both equity and debt ICS.

To conserve space, we briefly summarize the analysis and results here, and refer the reader to the Internet Appendix for details. We find that, for a reasonable calibration of the extended model, the debt ICS does not seem to play a quantitatively important role for the value and investment return spreads in this model. This result follows because the debt collateral constraint binds infrequently in the model, as firms endogenously increase the capital stock, reduce debt and/or save in cash to avoid hitting the binding constraint (a precautionary savings mechanism). As a result, the fraction of firms affected by the debt ICS is fairly small in equilibrium. Because the debt ICS only affects the investment decisions occasionally, it does not affect the dividend or the continuation value (price) of firms significantly. Given these effects, the endogenous cross-sectional risk dispersion generated by the debt ICS is fairly small in the extended model.

5.6.2 The empirical link between equity ICS, debt ICS, and asset prices.

Section 3.2 shows that the correlation of the equity ICS proxy with measures related to the cost of debt is low. Here, we examine the correlation of the equity ICS proxy with a proxy for the time-varying cost of issuing debt (debt ICS proxy). We construct this proxy as the first difference in the fraction of firms issuing debt in the cross-section, and we interpret the time-series variation in this measure as driven, at least partially, by changes in the cost of issuing debt.¹⁶ This approach is analogous to the estimation of the model-implied equity ICS proxy, because the changes in the fraction of firms issuing equity in the cross-section is the main explanatory variable of the variation in the true cost of issuing equity in the model (as reported in Table 3). (Orthogonalizing the debt ICS proxy to measures of the cost of debt, such as credit spread or risk-free rate, which are naturally relevant determinants of firm's debt issuance decision, does not change any of the qualitative analyses discussed here.)

The Internet Appendix shows that the correlation between the equity and debt ICS proxies is close to zero (3%), which reinforces the interpretation that the equity ICS proxy captures a dimension of variation in external financing conditions, that is, at least partially, specific to the equity, not the debt, market. For example, this low correlation is consistent with the interpretation that variation in equity issuance costs is driven by changes in asymmetric information because it is well known the asymmetric information is significantly more severe for equity than for debt (i.e., equity is more "informationally sensitive" than debt, in the sense of Dang, Gorton, and Holmström 2015; Gorton 2016), because (non-speculative-grade) debt can be transacted without much concern for adverse selection. Our results suggest that this different component driving equity issuance, but not debt issuance, seems to matter for asset prices.

In addition, the Internet Appendix replicates the asset pricing tests reported in Section 4 using the debt ICS proxy instead of the equity ICS proxy as the pricing factor in Equation (21). We find that the debt ICS proxy does not price well the set of portfolios considered here.

The results reported here suggest that the link between equity returns and variation in the cost of issuance of alternative sources of external financing is stronger for equity than for debt (at least for the low risk type of debt considered here).¹⁷ Our findings also highlight the importance of studying separately the

¹⁶ We define a firm as a debt issuer if the change in its total debt, including short-term debt (annual Compustat item DLC) and long-term debt (annual Compustat item DLTT), is positive.

¹⁷ Here, we are examining the cost of issuing debt, not necessarily the price of debt. Consistent with Jaganathan and Wang (1996), we find that the change in credit spread (return on debt), has explanatory power for book-to-market portfolios (when portfolio returns are equal weighted). However, it has no explanatory power for investment portfolios also considered in the paper. See the Internet Appendix for this analysis.

effects of time-varying equity issuance costs from the effects of time-varying debt issuance costs.¹⁸

6. Conclusion

We develop a dynamic model economy to evaluate the impact of time variation in the firm's ability to raise external equity financing on asset prices, real quantities, and financial flows in the cross-section of U.S. firms. The economic mechanism in the model emphasizes that the firms' ability to substitute between different marginal sources of external financing (equity and debt) during bad economic times is an important determinant of equilibrium risk premiums. In the model, growth and high investment firms can substitute more easily debt financing for equity financing when it becomes more costly to raise external equity, which tend to occur at times when marginal utility is high (bad economic times). Hence, these firms are less risky, and hence have relatively lower average returns in equilibrium.

We use the structure of the model to estimate an empirical proxy of aggregate equity ICS in the U.S. economy from 1964 to 2013, and provide empirical support for the models' assumptions and economic mechanisms. Consistent with the model, we show that the equity ICS captures systematic risk in the economy, and has a positive price of risk (marginal utility is high during times when is more costly for firms to issue equity). Thus, assets with returns that covary less with equity ICS (growth and high investment firms) have lower risk, and hence lower average returns in equilibrium. In addition, consistent with the model's economic mechanism, low risk growth and high investment firms (but not high risk value and low investment firms) issue more debt when the cost of equity financing increases. That is, the debt-equity substitution mechanism in the model has empirical support.

The results reported here have implications for asset pricing, corporate finance, and macroeconomics literature. The findings suggest that time variation in the aggregate cost of external equity financing has a significant impact on asset prices, real quantities, and financial flows in the cross-section. By affecting firms' investment and financing decisions, these shocks are likely to affect aggregate quantities as well. Thus, going forward, incorporating aggregate shocks to the cost of external equity financing in current dynamic stochastic general equilibrium (DSGE) models may be important for an accurate understanding of aggregate quantity dynamics, time-varying risk premiums, and financial flows over the business cycle.

Finally, in our analysis, we treat the aggregate equity ICS as exogenous, as a first step towards understanding the joint behavior of external equity financing

¹⁸ We note that this analysis focuses on the link between equity/debt issuance cost shocks and asset prices in the cross-section. Thus, our analysis is silent on the effects of these shocks on aggregate asset prices and aggregate quantities. It is possible that the debt issuance cost shocks have important implications for these aggregate-level variables, consistent with the analysis in Jermann and Quadrini (2012).

frictions, asset prices, and financial flows in the cross-section. To help us better understand the links between the financial sector and the real economy, future research should endogenize the source of the equity ICS in a DSGE model.

A-1. Stationary Representation of the Model

It is easy to verify that all variables grow with X_t on the balanced growth path. Define

$$\{V_t, D_t, E_t, Y_t, K_t, B_t, I_t, H_t, G_t, \Phi_t, \Psi_t\} = \{v_t X_t, d_t X_t, e_t X_t, y_t X_t, k_t X_{t-1}, b_t X_{t-1}, i_t X_t, h_t X_t, g_t X_t, \phi_t X_t, \psi_t X_t\}, \quad (\text{A1})$$

where $\{v_t, d_t, e_t, y_t, k_t, b_t, i_t, h_t, g_t, \phi_t, \psi_t\}$ are detrended stationary variables.

The stationary optimization problem can be written as follows:

$$v(\Delta x_t, z_t, \xi_t, k_t, b_t) = \max_{i_t, b_{t+1}, n_t} d_t + \mathbb{E}_t \left[M_{t,t+1} \frac{X_{t+1}}{X_t} v(\Delta x_{t+1}, z_{t+1}, \xi_{t+1}, k_{t+1}, b_{t+1}) \right] \quad (\text{A2})$$

$$s.t. \quad d_t = e_t - \psi_t \quad (\text{A3})$$

$$h_t = \max(-e_t, 0) \quad (\text{A4})$$

$$e_t = (1 - \tau)y_t + \tau \delta k_t \frac{X_{t-1}}{X_t} + \tau r_f b_t \frac{X_{t-1}}{X_t} \mathbf{1}_{\{b_t \geq 0\}} \quad (\text{A5})$$

$$-i_t - g_t + b_{t+1} - (1 + r_l)b_t \frac{X_{t-1}}{X_t} - \phi_t \quad (\text{A6})$$

$$k_{t+1} = (1 - \delta)k_t \frac{X_{t-1}}{X_t} + i_t \quad (\text{A7})$$

$$b_{t+1} \leq \varphi k_{t+1}, \quad (\text{A8})$$

where the stationary output and various adjustment costs are given as follows:

$$y_t = Z_t \left(\frac{X_t}{X_{t-1}} \right)^{-\theta} k_t^\theta, \quad (\text{A9})$$

$$g_t = \begin{cases} \frac{c_k^+}{2} \left(\frac{i_t}{k_t} \right)^2 k_t \frac{X_t}{X_{t-1}}, & i_t \geq 0 \\ \frac{c_k^-}{2} \left(\frac{i_t}{k_t} \right)^2 k_t \frac{X_t}{X_{t-1}}, & i_t < 0 \end{cases}, \quad (\text{A10})$$

$$\phi_t = \frac{c_b}{2} \left(\frac{\Delta b_t}{b_t} \right)^2 b_t \frac{X_t}{X_{t-1}}, \quad (\text{A11})$$

$$\psi_t = \left[\eta_0 + \eta_1 h_t + \eta_2 \frac{h_t^2}{k_t} \frac{X_t}{X_{t-1}} \right] \exp[-\eta_3 \xi_t] \mathbf{1}_{\{h_t > 0\}}, \quad (\text{A12})$$

where $\Delta b_t = b_{t+1} - b_t \frac{X_{t-1}}{X_t}$.

Finally, the stock return is given as follows:

$$R_{t+1} = \frac{V_{t+1}}{V_t - D_t} = \frac{v_{t+1} \frac{X_{t+1}}{X_t}}{v_t - d_t}. \quad (\text{A13})$$

A-2. Numerical Algorithm

To solve the model numerically, we use the value function iteration procedure to solve the firm's maximization problem. The value function and the optimal decision rule are solved on a grid in a discrete state space. We specify two grids of 42 points for capital and 37 points for debt/cash, respectively, with upper bounds $\bar{k}$ and $\bar{b}$ that are large enough to be nonbinding. The grids for capital and debt/cash are constructed recursively, following Livdan, Saprizza, and Zhang (2009), that is, $k_i = k_{i-1} + c_{k1} \exp(c_{k2}(i - 2))$, where $i = 1, \dots, n$ is the index of grids points and c_{k1} and c_{k2} are two constants chosen to provide the desired number of grid points and two upper bounds $\bar{k}$ and $\bar{b}$, given two pre-specified lower bounds $\underline{k}$ and $\underline{b}$. The advantage of this recursive construction is that more grid points are assigned around $\underline{k}$ and $\underline{b}$, where the value function has most of its curvature.

The aggregate productivity shock ε_t^x is an i.i.d. standard normal shock. We discretize ε_t^x into 3 grid points using Gauss-Hermite quadrature. The state variables ξ and z have continuous support in the theoretical model, but they have to be transformed into discrete state space for the numerical implementation. The popular method of Tauchen and Hussey (1991) does not work well when the persistence level is above 0.9. Because both the aggregate issuance cost wedge ξ and idiosyncratic productivity process z are highly persistent, we use the method described in Rouwenhorst (1995) for a quadrature of the Gaussian shocks. We use 7 grid points for the ξ process and 5 grid points for the z process. In all cases, the results are robust to finer grids as well. Once the discrete state space is available, the conditional expectation can be carried out simply as a matrix multiplication. Spline interpolation is used extensively to obtain optimal investment and debt that do not lie directly on the grid points. Finally, we use a simple discrete global search routine in maximizing the firm's problem.

A-3. Data Definitions and Portfolio Construction

Monthly stock returns are from the Center for Research in Security Prices (CRSP), and accounting information is from the CRSP/Compustat Merged Annual Industrial Files. The sample is from 1964 to 2013, and includes firms with common shares (shrcd=10 and 11) and firms traded on NYSE, AMEX, and NASDAQ (exchcd=1, 2, and 3).¹⁹ We omit firms whose primary standard

¹⁹ The sample starts in 1964 due to the availability of the data items required to construct the portfolio sorting variables (book equity and investment rate).

industry classification (SIC) code is between 4900 and 4999 (utility firms) or between 6000 and 6999 (financial firms). We correct for the delisting bias following the approach in Shumway (1997).

We construct the following firm characteristics. Following Belo, Lin, and Bazdresch (2014), the firm's investment in physical capital rate (IK_t) is given by $IK_t = I_t / (0.5 \times (K_{t-1} + K_t))$, in which investment is Compustat data item CAPX (capital expenditures) minus Compustat data item SPPE (sales of property, plant, and equipment), and the physical capital stock (K_t) is given by Compustat data item PPENT (net property plant and equipment). Missing values of SPPE are set to zero. BM is the book equity to market equity ratio, where both book equity and market equity values follow the definitions in Fama and French (1992). Market leverage ratio is computed as book value of debt (DLTT + DLC) over the sum of market value of equity and book value of debt. Book leverage ratio is computed as book value of debt over total asset (AT). The gross equity issuance of a firm equals to Compustat data item SSTK if SSTK is greater than 3% of its average year-begin and year-end market equity. The equity issuance fraction of a portfolio is defined as the number of issuing firms over the total number of firms for a year. A firm's profitability is defined as (SALES - COGS) / AT. Firm-level TFP is the total factor productivity estimated in İmrohoroğlu and Tüzel (2014).

We construct the five book-to-market portfolios and five investment rate portfolios as follows. Following Fama and French (1993), we sort the portfolios annually. At the end of June of year t , we sort all stocks in our sample by either the firms' book-to-market ratio or investment rate in year $t - 1$. The portfolio breakpoints to allocate firms into portfolios are set as the 20th, 40th, 60th, and 80th percentiles of the cross-sectional distribution of the sorting variable across NYSE firms. The firms are then allocated across the portfolios based on their sorting variable relative to the breakpoints. Then, from July of year t to June of year $t + 1$, we compute both equal- and value-weighted monthly portfolio returns across all the firms in the portfolio using market equity in the end of the previous month as the weights. The monthly portfolio returns are then transformed to annual (from January to December) by compounding the monthly returns. Portfolio excess returns are obtained after subtracting the annual risk-free rate. For the five investment rate portfolios we also require firms to have December fiscal year end (annual Compustat item FYR), because of the relatively low persistence of the investment-rate relative to book-to-market and size characteristic, in which case the lag between the accounting data and the sorting procedure introduces substantial measurement error in the allocation of firms across the portfolios (see Belo, Lin, and Bazdresch (2014) for a discussion of this issue). Relaxing this data requirement and performing the sorting procedure at the monthly frequency using the most up to date accounting data (maintaining at least a four month lag between the accounting data and the portfolio procedure) produces portfolio characteristics that are very similar to those reported here with the annual sort and fiscal year end restriction.

We construct the 5×5 size and book-to-market portfolios and the 5×5 size and investment rate portfolios using a similar procedure. Firm size is measured by market equity. Following Fama and French (1993), the double sorted portfolios are the intersections of five portfolios formed on size and five portfolios formed on book-to-market ratio or investment rate. The portfolio breakpoints to allocate firms into portfolios are set as the quintiles of each sorting variable across NYSE firms. For the 5×5 size and investment rate portfolios, we also require firms to have a December fiscal year-end.

Industry portfolios, the stock market factor and the risk-free returns are obtained from Kenneth French's website. Portfolio returns are annualized (compounding the monthly returns) from January to December, to be consistent with the frequency of the equity issuance data.

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